

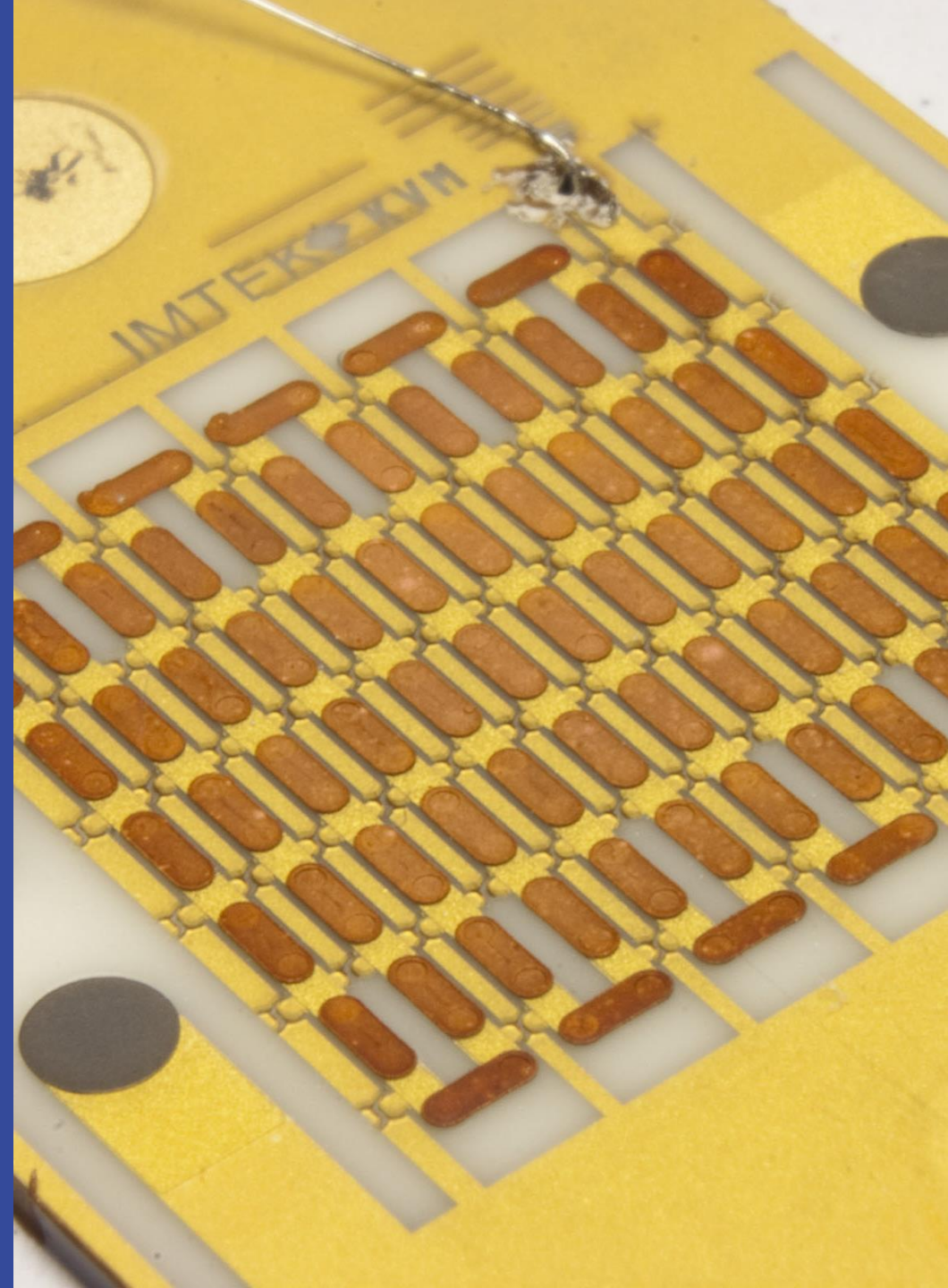
Micro Energy Harvesting

Chapter 3: Thermoelectric Energy Harvesting

Micro Energy Harvesting SoSe 2026

IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

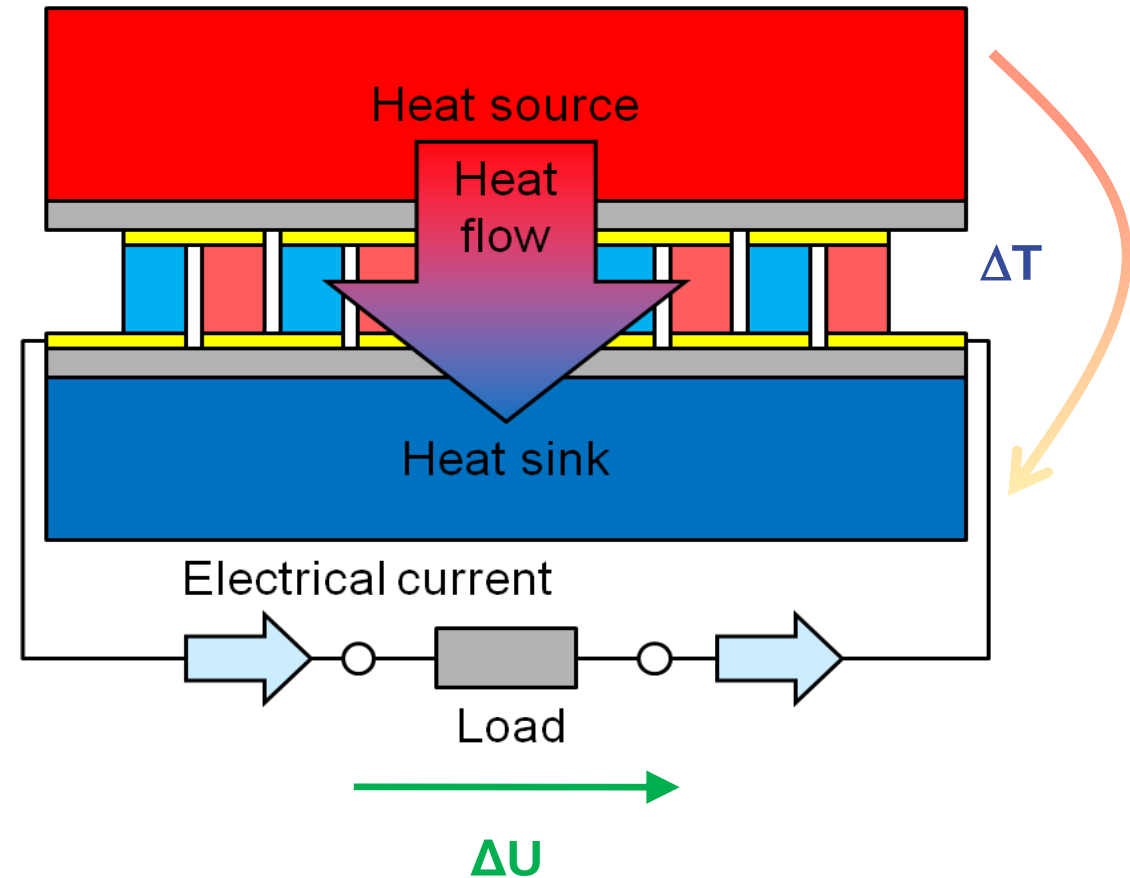
Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank Goldschmidtboing



Thermoelectric Generator - TEG

Principle of operation

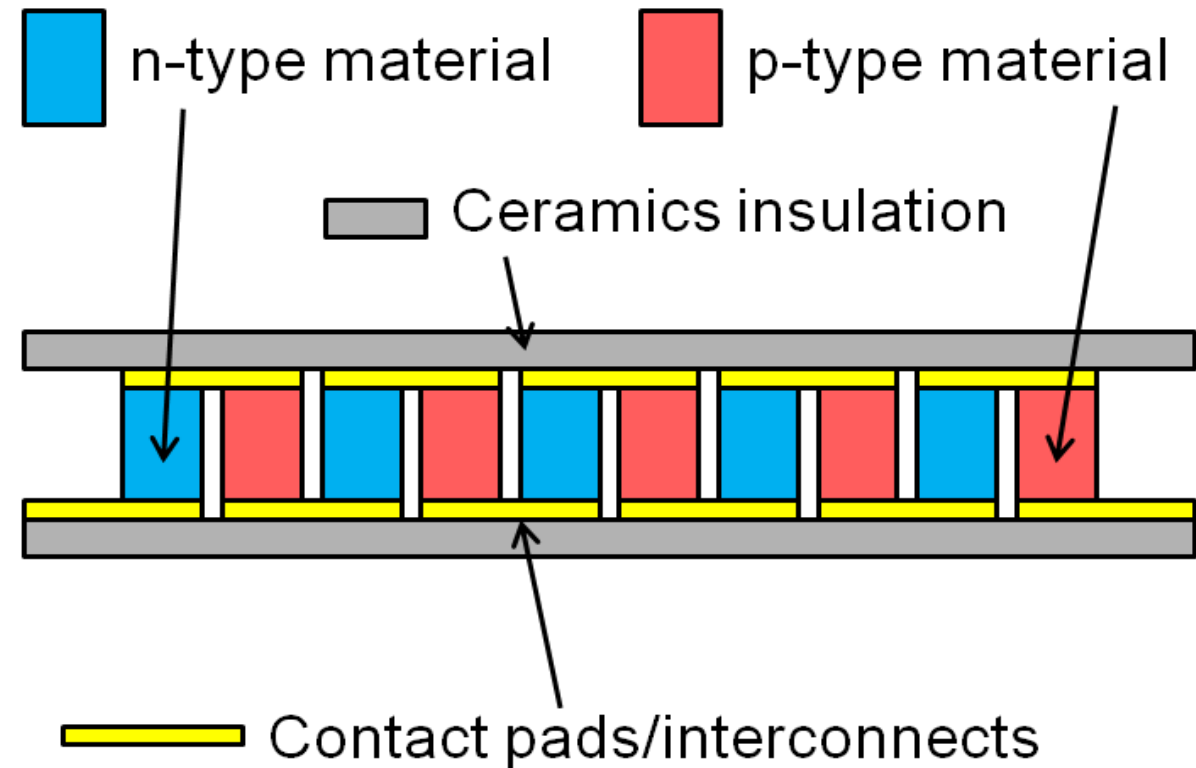
- TEG attached to **heat source** and **heat sink**
- TEG converts **thermal** into **electrical energy**, for that
- a **heat flow** is (and must be) created by the temperature difference ΔT across the device



Thermoelectric Generator

Components of a “standard” TEG

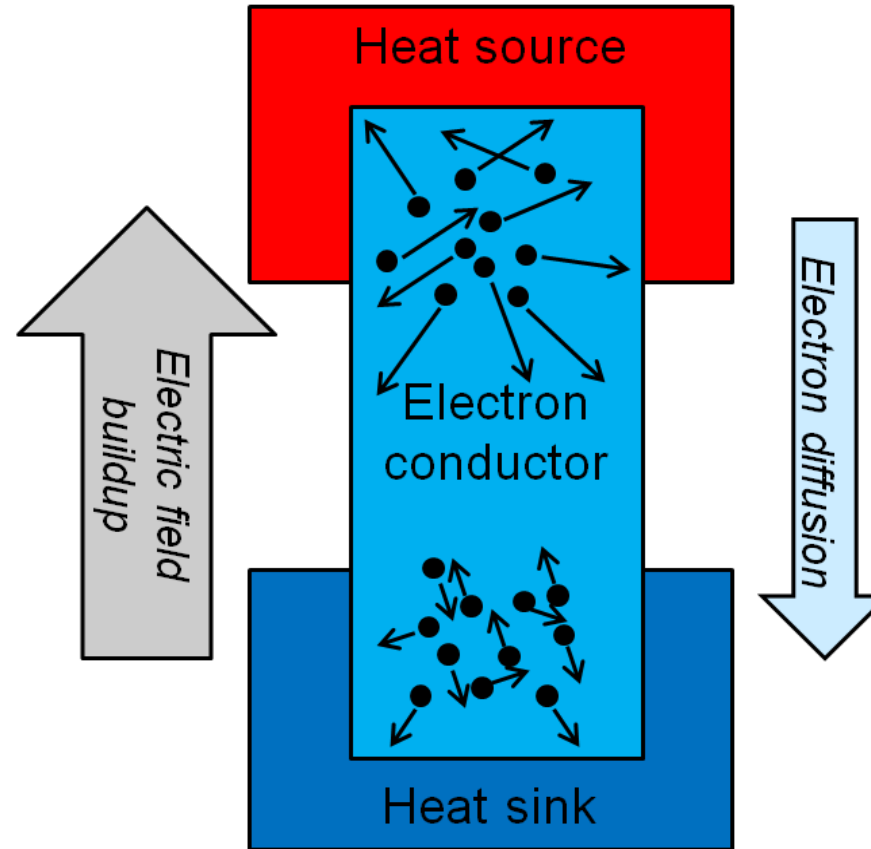
- **n- and p-type materials**
thermoelectrically active components (“thermo-legs”)
- **interconnects**
electrical connection between thermo-legs
- **contact pads**
electrical connection to user application
- **ceramics insulation**
electrical insulation, mechanical stability



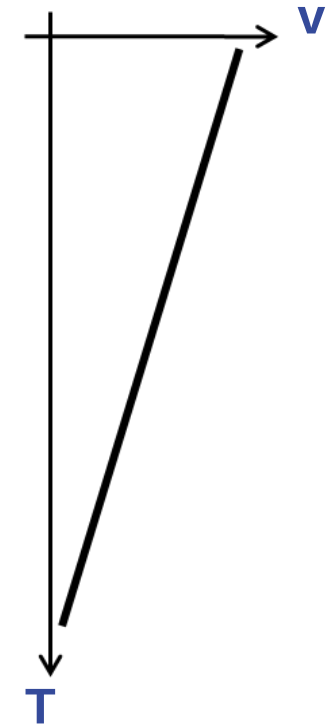
The Seebeck Effect

Thermodiffusion

- higher average electron velocity at higher temperature (“heat source”)
- electrons diffuse towards cold side (“heat sink”)
- equilibrium is reached through build-up of a counteracting electric field



average electron velocity



The Seebeck Effect

Seebeck Coefficient / Voltage

- Typical values on the order of a few 10s of microvolts per Kelvin
- Small for metals, higher values for semiconductors

Unfortunately two different scales are in use:

- “Old” one, with platinum as a reference material with a value of 0 $\mu\text{V/K}$
- “New” one, with absolute values of the Seebeck coefficient (Platinum at $\sim -5 \mu\text{V/K}$)

Metals	Seebeck Coefficient	Semiconductors	Seebeck Coefficient
	$\mu\text{V/K}$		$\mu\text{V/K}$
Antimony	47	Se	900
Nichrome	25	Te	500
Molybdenum	10	Si	440
Cadmium	7.5	Ge	300
Tungsten	7.5	n-type Bi_2Te_3	-230
Gold	6.5	p-type $\text{Bi}_{2-x}\text{Sb}_x\text{Te}_3$	300
Silver	6.5	p-type Sb_2Te_3	185
Copper	6.5	PbTe	-180
Rhodium	6.0	$\text{Pb}_{0.3}\text{Ge}_{3.9}\text{Se}_{5.8}$	1670
Tantalum	4.5	$\text{Pb}_{0.6}\text{Ge}_{3.6}\text{Se}_{5.8}$	1410
Lead	4.0	$\text{Pb}_{0.9}\text{Ge}_{3.3}\text{Se}_{5.8}$	-1360
Aluminum	3.5	$\text{Pb}_{1.3}\text{Ge}_{2.9}\text{Se}_{5.8}$	-1710
Carbon	3.0	$\text{Pb}_{1.5}\text{Ge}_{3.7}\text{Se}_{5.8}$	-1990
Mercury	0.6	SnSb_4Te_7	25
Platinum	0	SnBi_4Te_7	120
Sodium	-2.0	$\text{SnBi}_3\text{Sb}_1\text{Te}_7$	151
Potassium	-9.0	$\text{SnBi}_{2.5}\text{Sb}_{1.5}\text{Te}_7$	110
Nickel	-15	$\text{SnBi}_2\text{Sb}_2\text{Te}_7$	90
Constantan	-35	PbBi_4Te_7	-53
Bismuth	-72		

<http://www.electronics-cooling.com>

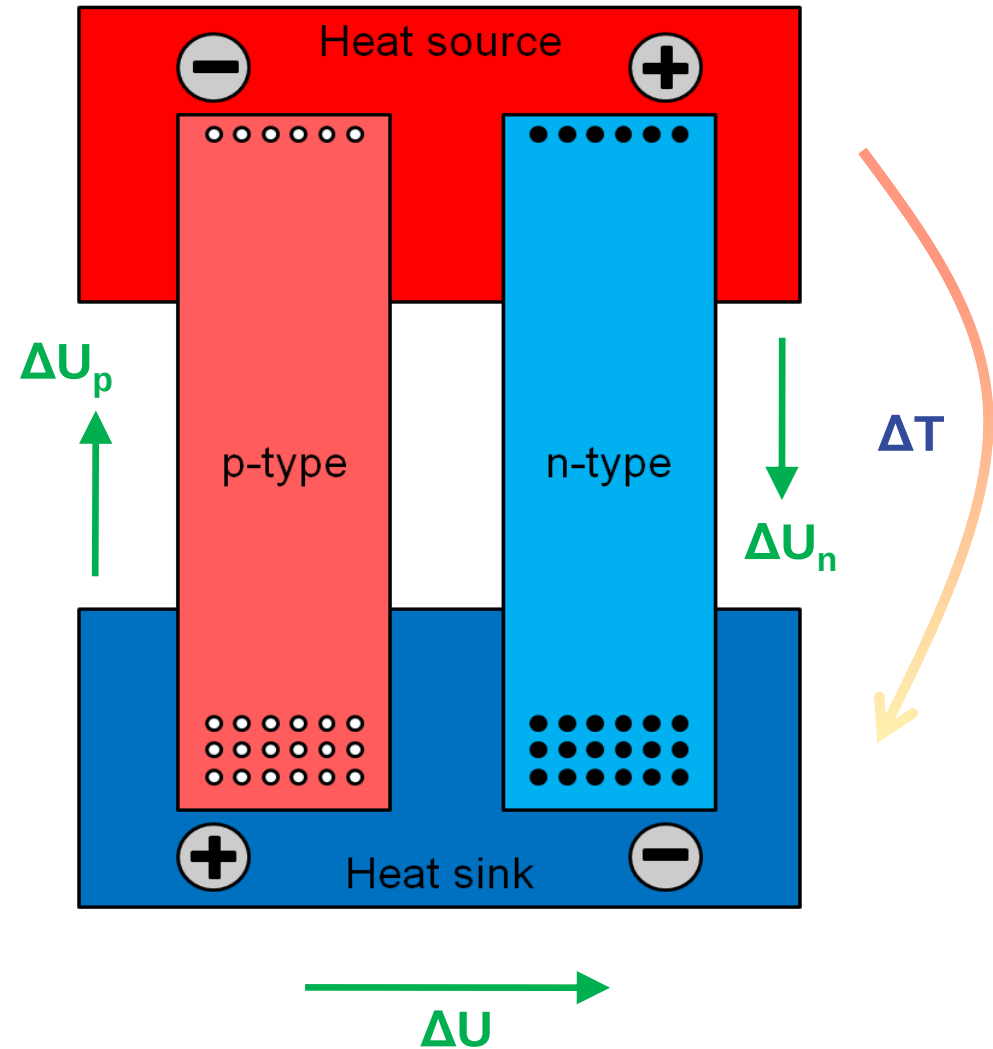
The Seebeck Effect

Thermopower

- Resulting electric field has different direction for p- and n-type materials
- Positive (+)S for p-type and negative (-)S for n-type materials
- Magnitude depends on temperature difference and material parameters

$$\begin{aligned}\Delta U &= \Delta U_p + \Delta U_n \\ &= (S_p - S_n) \cdot \Delta T = S \cdot \Delta T\end{aligned}$$

*S (or α) is the Seebeck coefficient
(or thermopower) of a material*

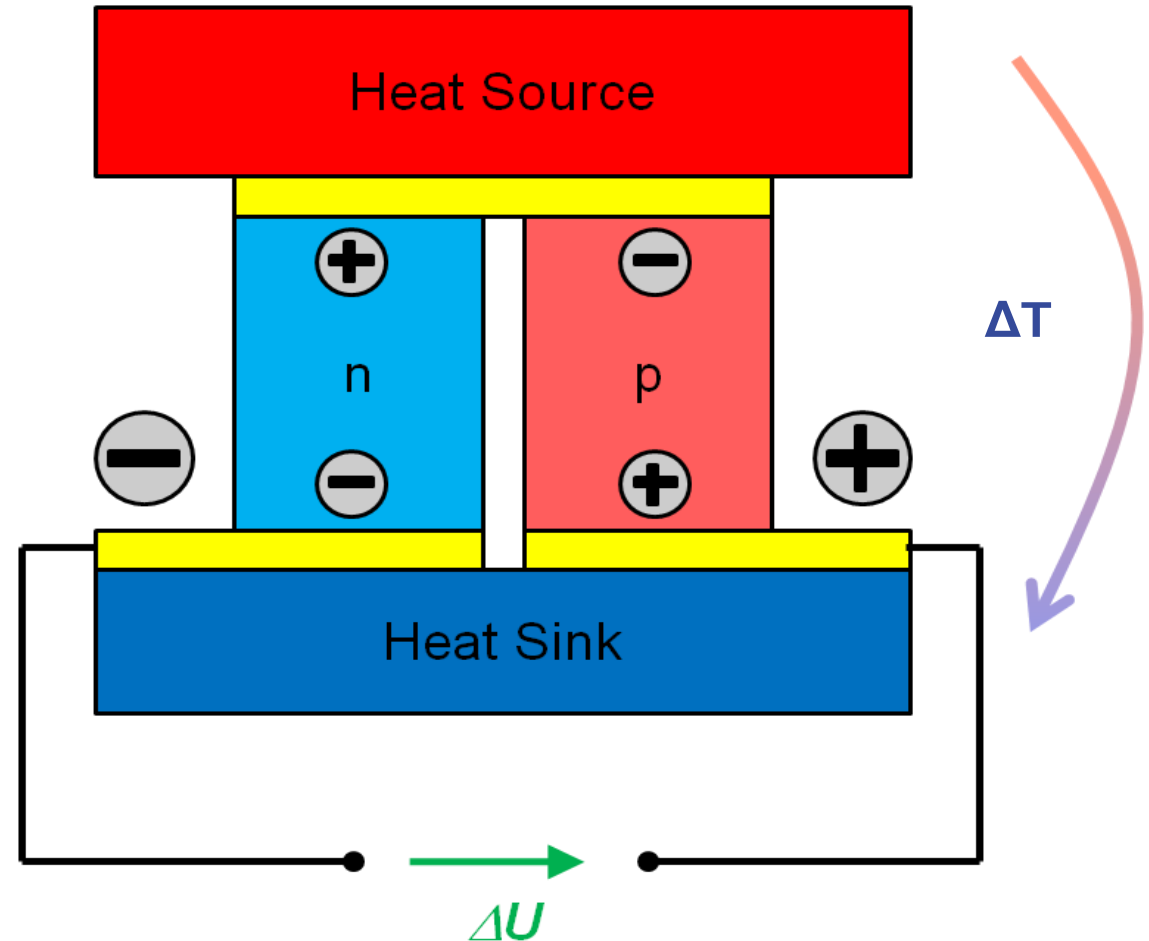


The Seebeck Effect

Thermoelectric Unicouple

- unit cell of every TEG
- n- and p-type materials in parallel to allow a simple unidirectional heat flow
- Typical voltage generated:
 $\Delta U = 200\text{-}500 \mu\text{V/K}$
- Thermo-voltages added by connecting thermoelegs electrically in series
- Large numbers of thermoelegs for a useful output voltage

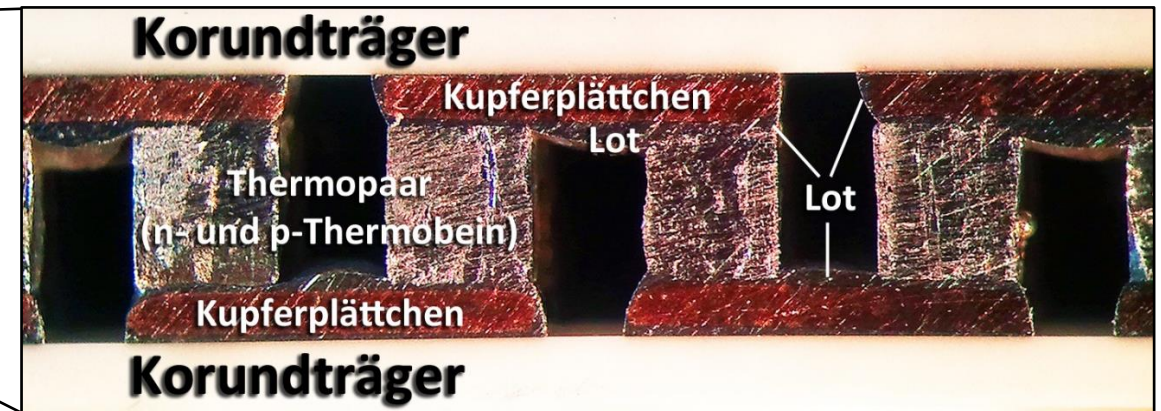
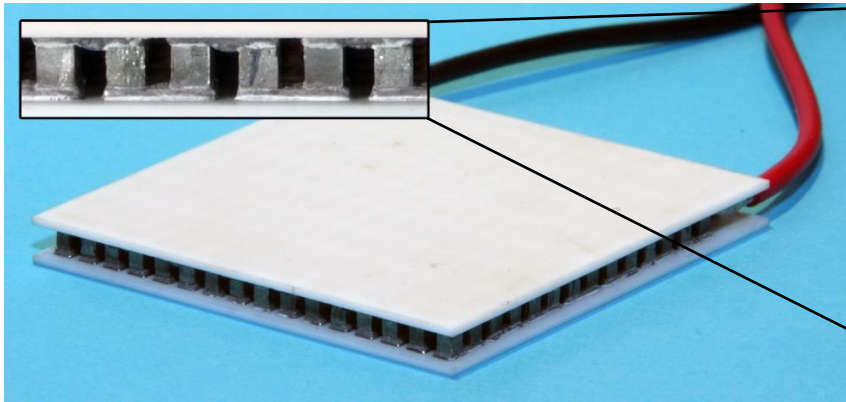
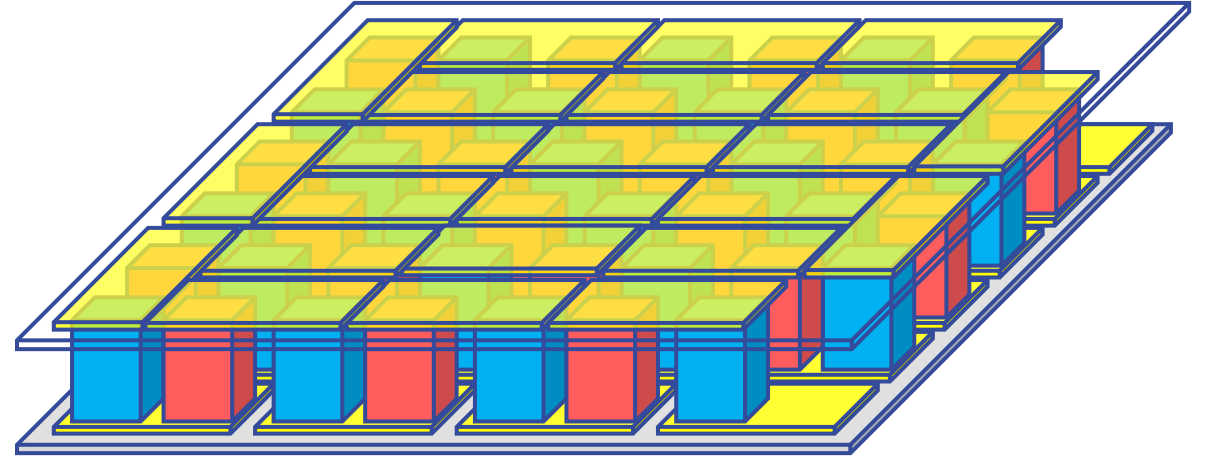
$$\Delta U = S \cdot \Delta T$$



Thermoelectric Generators

Macroscopic Design

- large array of thermocouples, typically around 20 to 150 pairs
- Dimensions: $1 \times 1 \text{ cm}^2 \dots 6 \times 6 \text{ cm}^2$
- height: $0.3 \dots 1.5 \text{ cm}$
- voltage: $10 \dots 100 \text{ mV/K}$

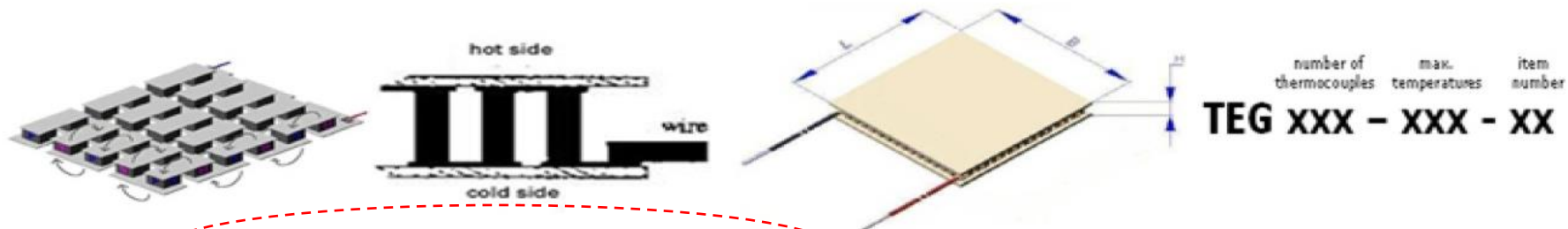


Thermoelectric Generators

Datasheets

Data of a typical „macroscopic“ TEG - Thermalforce TEG 049-150-30

THERMO ELECTRICAL GENERATOR



number of thermocouples max. temperatures item number
TEG xxx - xxx - xx

TEG 049-150-30 Angaben bei $\Delta T = 120K$ $T_{hot} = 150^\circ C$ und $T_{cold} = 30^\circ C$

Maße dimensions L x B x H mm	U U leer open circuit voltage V	U U last load voltage V	R R bei 300 K resistance at 300 K Ω	R R kr resistance corrected Ω	I I Kurzschl. short circuit current A	I I last load current A	Leistung W el. load power W	Thermokraft, Seebeckkoeff. V/K thermoforce at 300 K α	Thermokraft, Seebeckkoeff. V/K kr thermoforce corrected α	Wärmeleitwert W/K thermal conduct. κ	Wärmewiders K/W thermal resistance R_{th}	Wirkungs- grad efficiency η	aufgen. Wärme heat flow hot side Q hot	abzuführende Wärme heat flow cold side Q cold
25x25x3,7	2,054	1,027	0,570	0,805	2,551	1,275	1,310	0,016	0,018	0,190	5,263	4,256	-30,8	29,6
	Angaben bei $\Delta T = 100K$ $T_{hot} = 150^\circ C$ und $T_{cold} = 50^\circ C$													
	1,728	0,864	0,570	0,839	2,059	1,030	0,890	0,016	0,019	0,190	5,263	3,485	-25,5	24,8
	Angaben bei $\Delta T = 100K$ abzüglich Temperaturabfall durch die Keramikplatten (15 W/mK) und Wärmeleitpaste (10 W/mK) bei durchschnittl. therm. Anbindung													
	1,555	0,778	0,570	0,839	1,854	0,927	0,721	0,016	0,019	0,190	5,263	3,141	-22,9	22,3

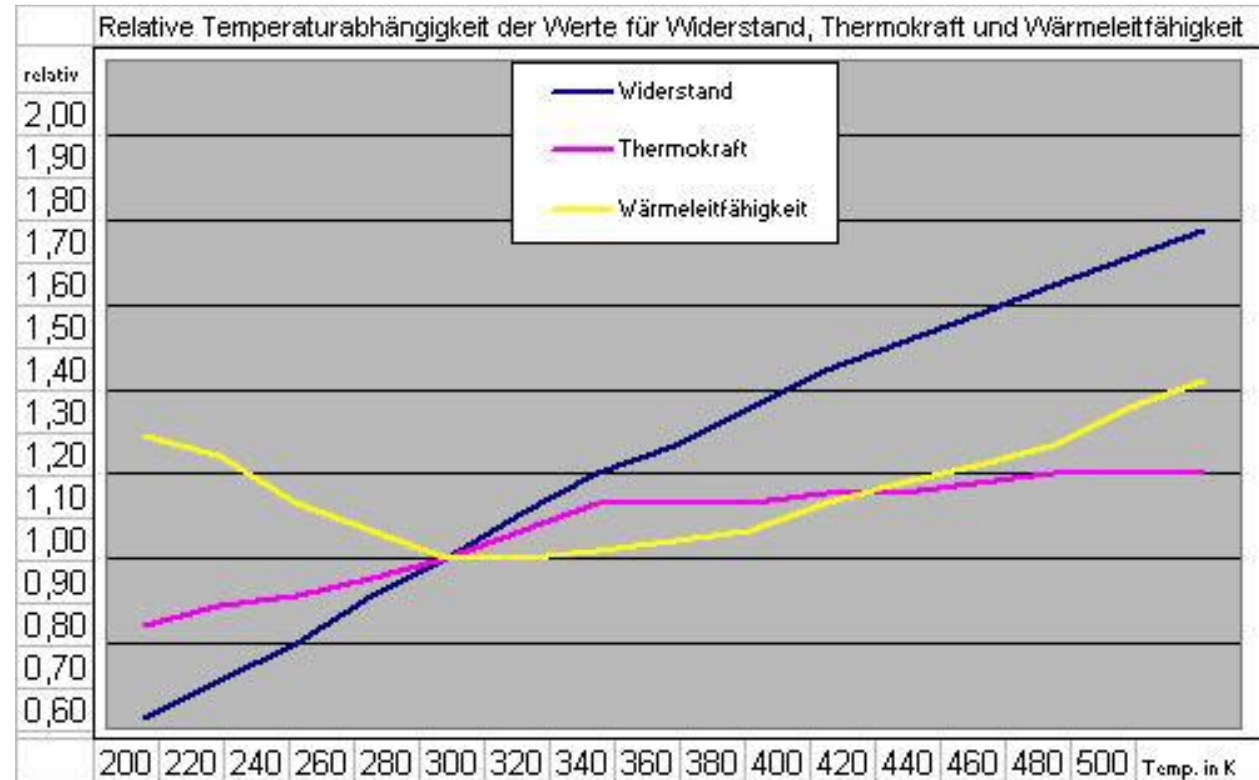
Thermoelectric Generators

Datasheets

Temperature influence on the most important parameters

General Trends

- increase of internal resistance with temperature
 - variations of thermal resistance with temperature
 - Increase of Seebeck coefficient with temperature
- ➡ power loss
 - ➡ power gain or loss
 - ➡ power gain



Thermoelectric Generators

Datasheets

*Other important
parameters
and requirements*

Mechanische Daten/mechanical data:

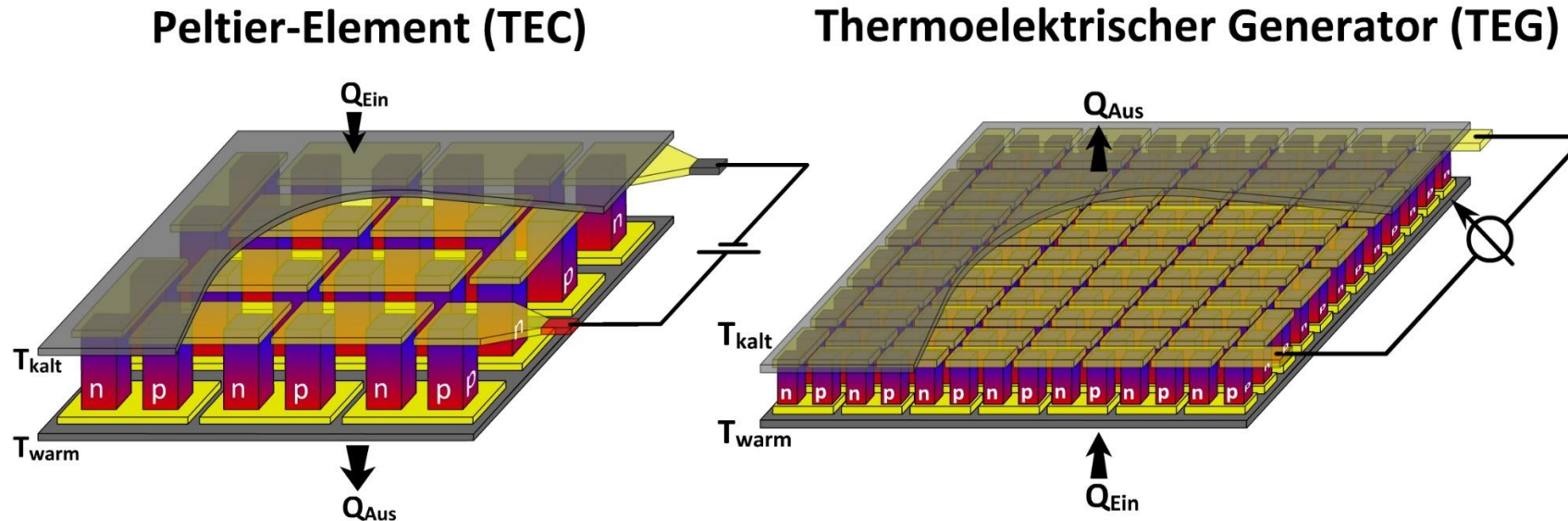
Anzahl/number of thermocouples:	049
Material/material:	Bi_2Te_3
Höhtoleranz/height tolerance:	$\Delta H \pm 0,25 \text{ mm}$
Längentoleranz/length tolerance:	0,2 mm
Gewicht/weight:	12 g
max. Temperatur/max. temperatures:	150 °C
max. Scherkraft/max. shear force:	pro cm^2 , 125 N
max. Vibration/max. vibration:	bis 55 Hz, 2 m/s^2
max. Zugkraft/ max. tension:	pro cm^2 , 195 N
Keramik Platten/ceramic plates:	BK-96, Al_2O_3 , 96 %
thermal conductivity of the ceramic:	15 W/mK, Höhe/height: 1 mm
Toleranz der thermischen und elektrischen Parameter:	$\pm 6 \%$
tolerances of thermal and electric parameters:	$\pm 6 \%$
Empfohlener Kühler/recommended cooler:	active air cooler, R_{th} : 0,58 K/W

Requirements

- flexible mounting concept (variations in height ΔH)
- protection against over temperature, shock, forces
- active cooling „recommended“

Thermoelectric Generators and Peltier Elements

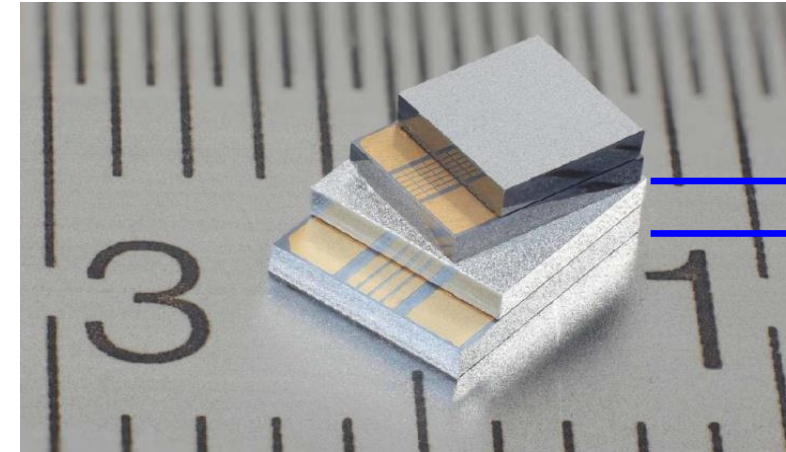
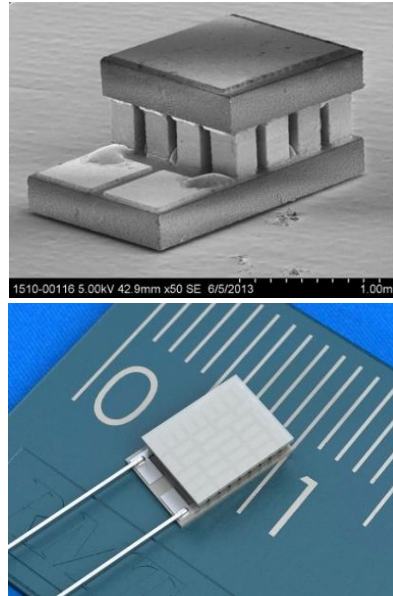
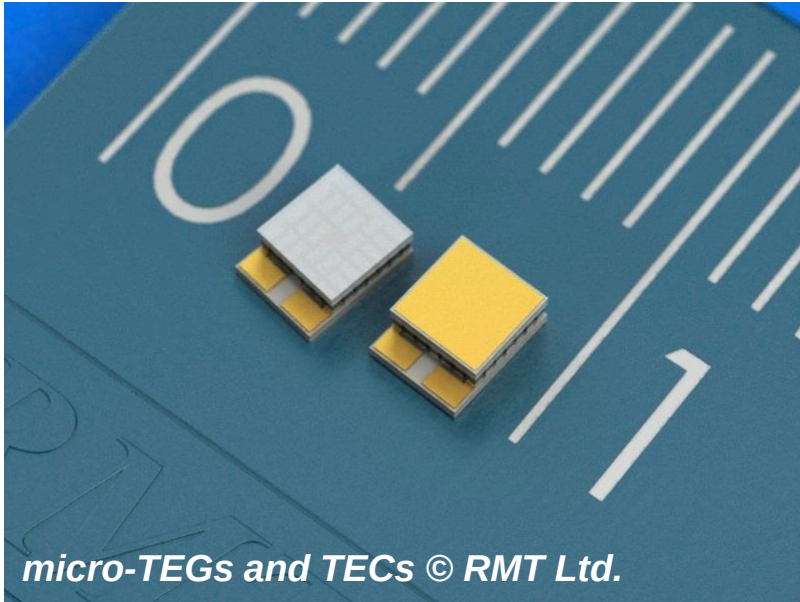
What's the difference?



TEC		TEG
max. cooling performance / max. thermal transport	<i>optimized for...</i>	max. output voltage / max. conversion efficiency
lower	<i>No. of TE-legs</i>	higher
larger	<i>TE-leg crossection</i>	smaller

Thermoelectric Microgenerators

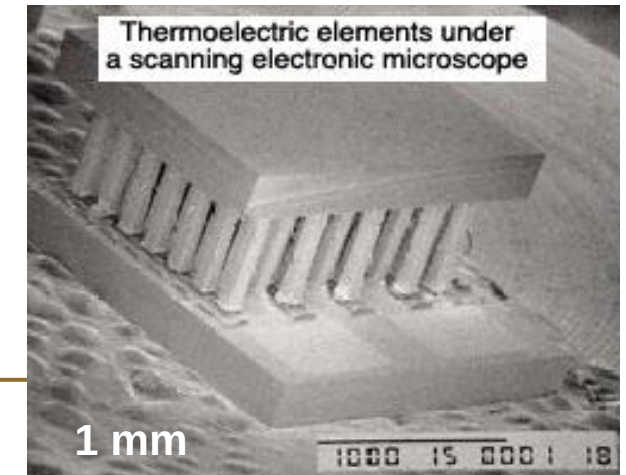
μ TEGs



1 mm
micro-TEG
© MicroPelt

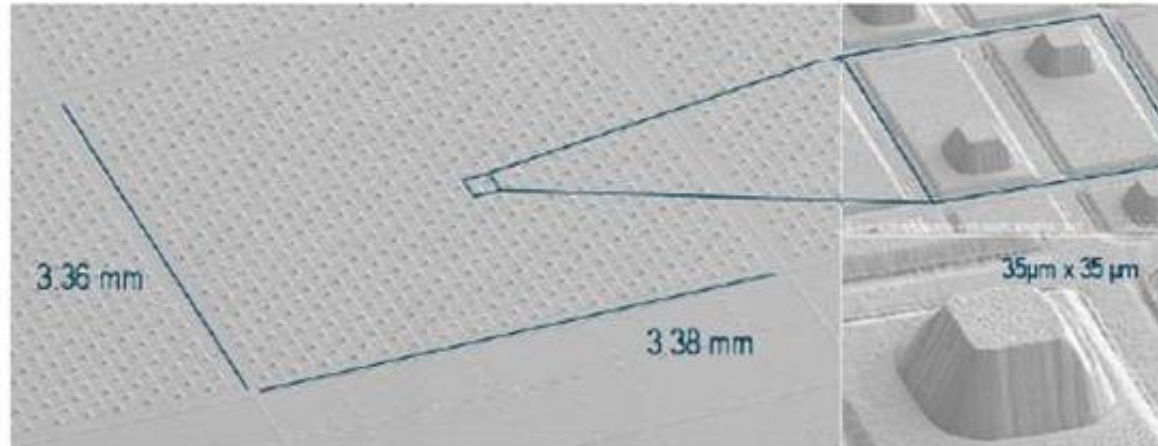
power: μ W to mW

„Seiko Thermic“ wristwatch (small series in 1998)
and micro-TEG developed by Seiko (1994)



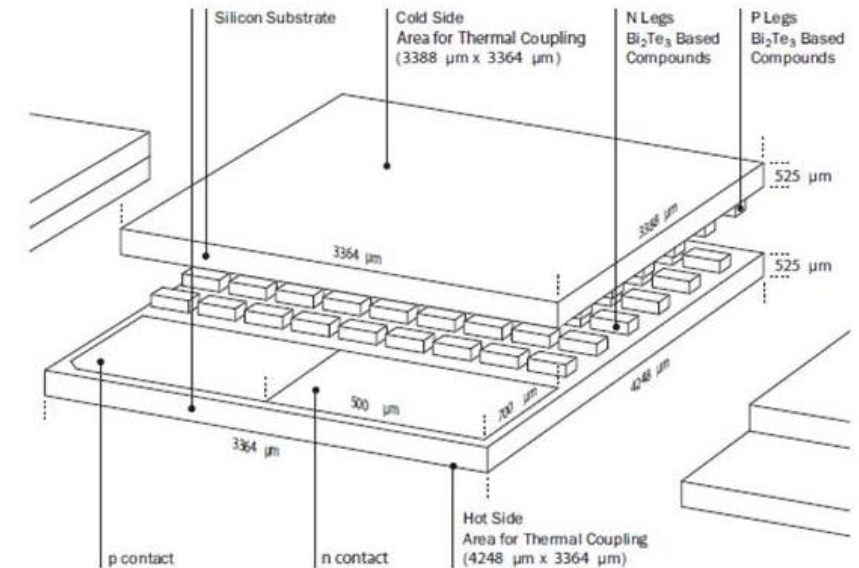
Thermoelectric Microgenerators

MicroPelt μ TEG



$A=35 \times 35 \mu\text{m}^2$

MPG-D751 dimensions (schematic drawing)



Type	Maximum Dimensions [mm] Top side Bottom side	Minimum Dimensions [mm] Top side Bottom side	Number of leg pairs	Thermal Resistance at 85°C	Electrical Resistance at 23°C	Net Seebeck Voltage at 23°C	Thickness [μm]
MPG-D651	2.5 x 2.5 3.375 x 2.5	2.45 x 2.45 3.325 x 2.45	286	22 K/W	185 Ω	75 mV/K	1090
MPG-D751	3.388 x 3.364 4.248 x 3.364	3.388 x 3.314 4.198 x 3.314	540	12.5 K/W	300 Ω	140 mV/K	1090

© MicroPelt

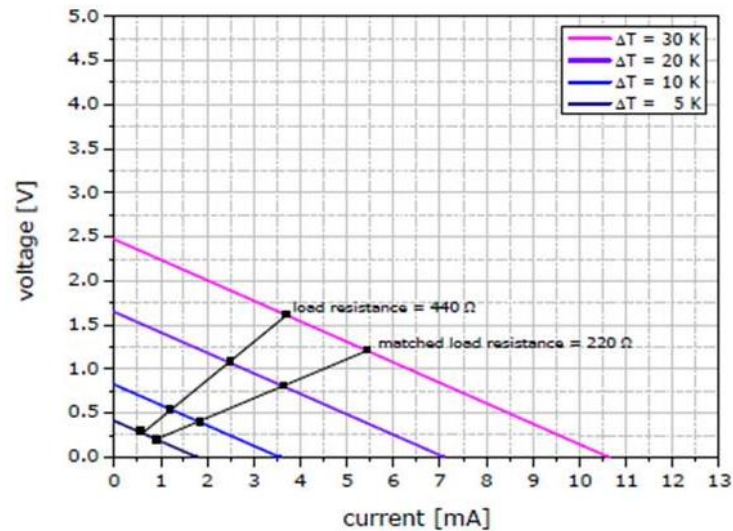
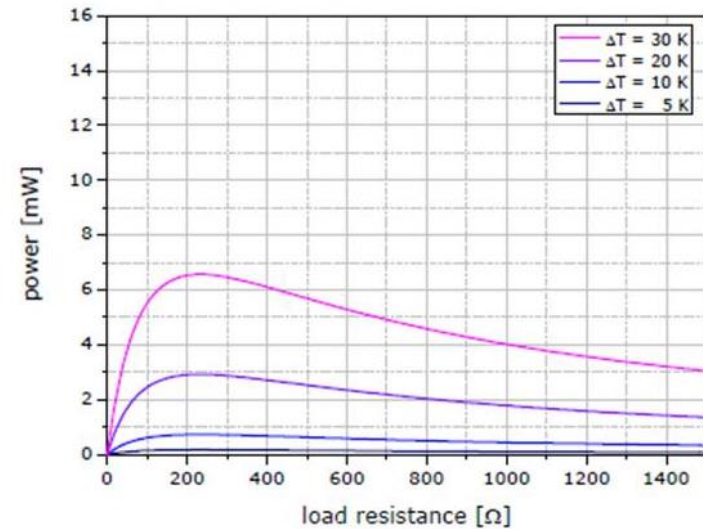
Thermoelectric Microgenerators

MicroPelt μ TEG

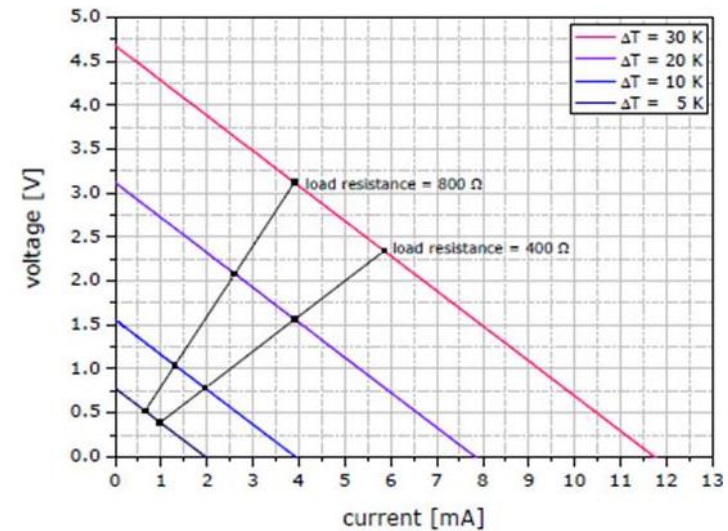
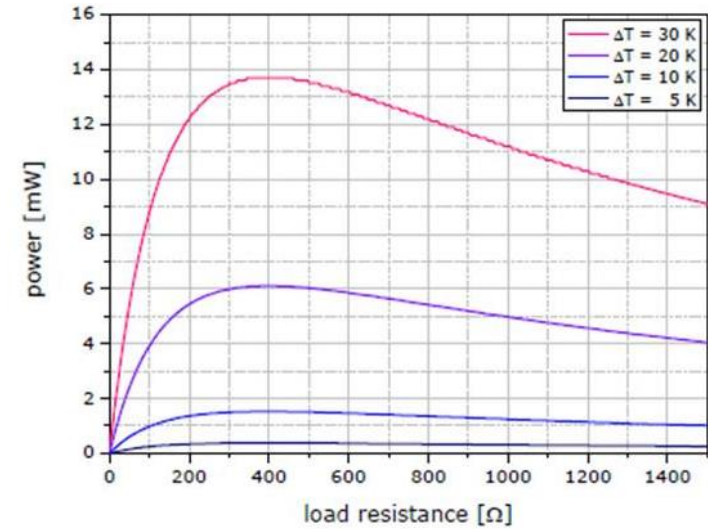
Power vs. load resistance

Voltage vs. Current at
different load resistances

MPG-D651



MPG-D751



© MicroPelt

MicroPelt μ TEG

Fabrication Process

Wafer A – n-type

Wafer B – p-type

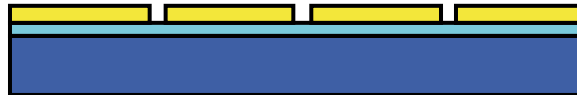
- Substrate: Si / SiO₂



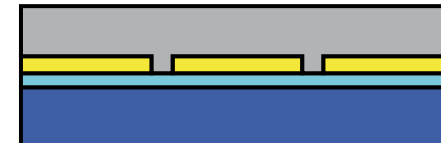
- Contact layer deposition



- Contact layer - Structuring



- Thermoelectric material deposition (sputtering)



MicroPelt μ TEG

Fabrication Process

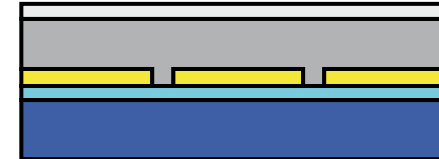
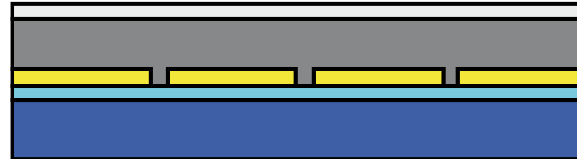
Wafer A – n-type

Wafer B – p-type

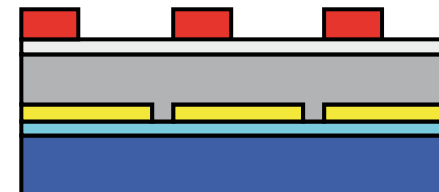
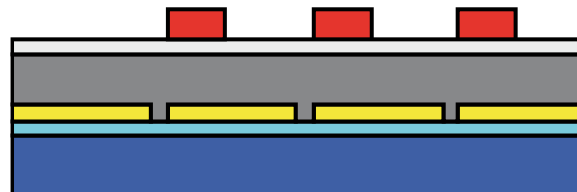
- Thermoelectric material deposited



- Deposition of solder layer



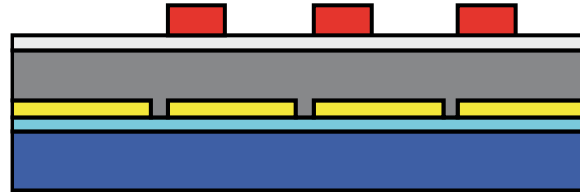
- Photoresist structuring of thermolegs



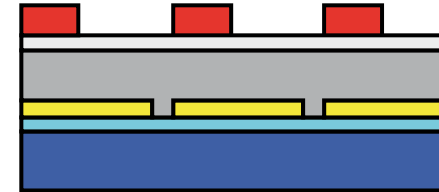
MicroPelt μ TEG

Fabrication Process

Wafer A – n-type

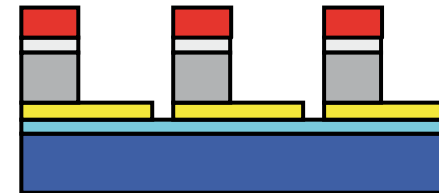
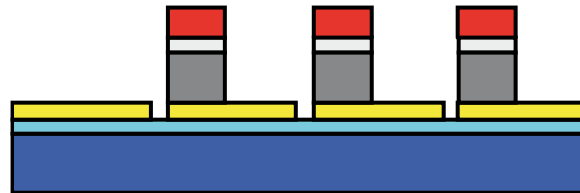


Wafer B – p-type

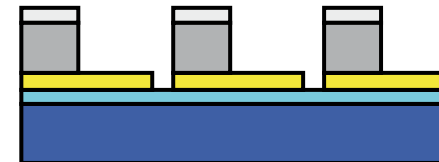
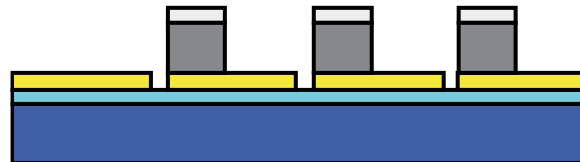


- Structured photoresist

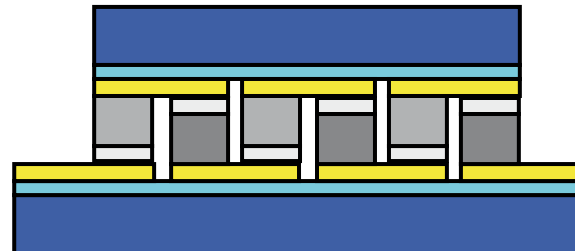
- Isotropic and selective etching (very challenging)



- Photoresist removal



- Flip-Chip bonding process



Thermoelectric Microgenerators

Different Approaches

- **MicroPelt**

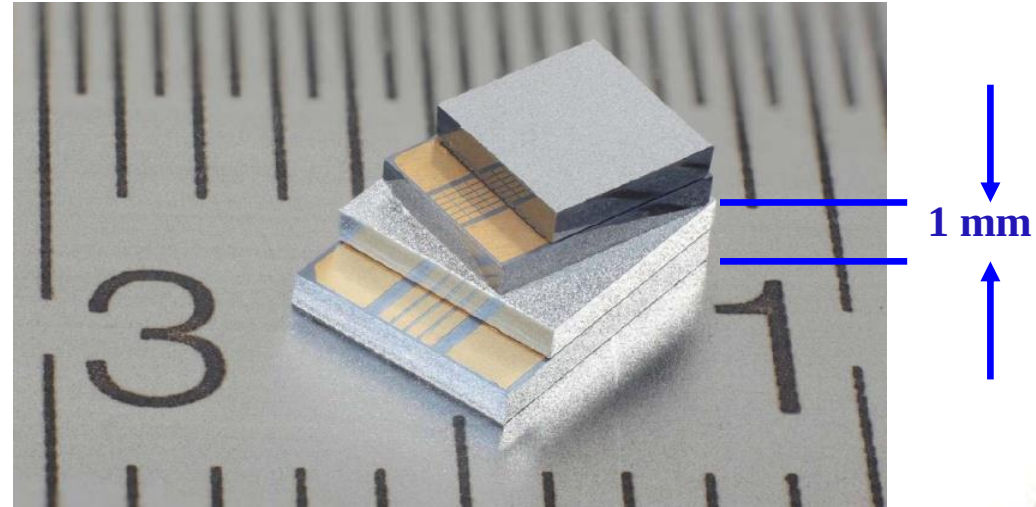
Sputter deposition on Si-Substrate

- **greenTEG**

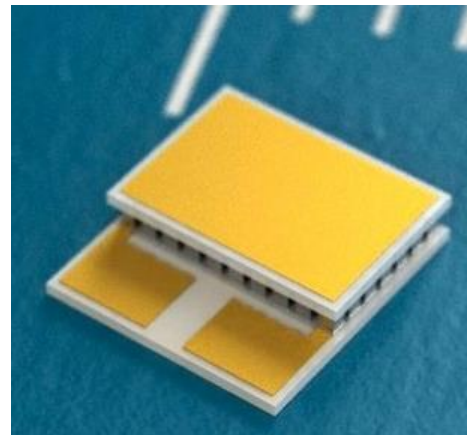
Electrodeposition of thermoelectric material

- **RMT Ltd.**

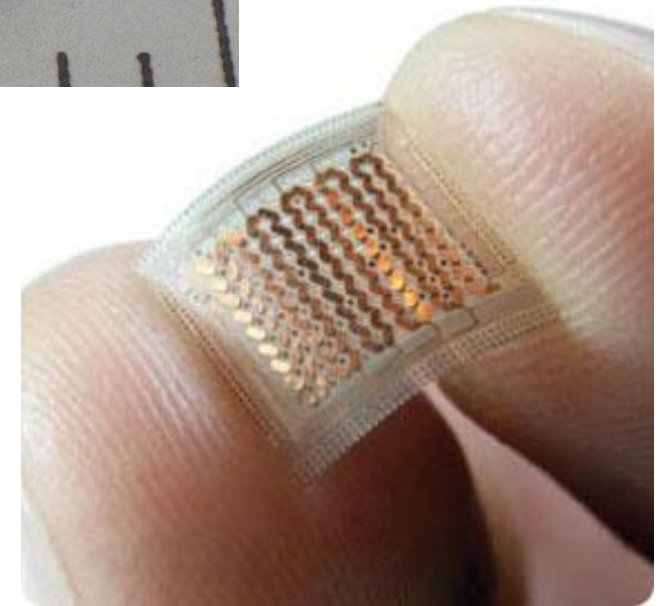
Microstructured, pellet-based classical design



© MicroPelt



© RMT Ltd.



© greenTEG, Zürich

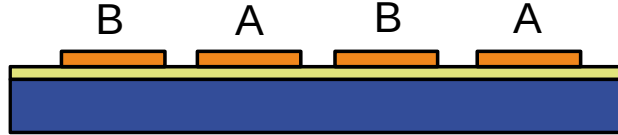
greenTEG

Fabrication Process

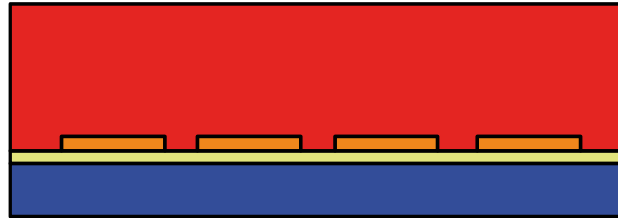
- Substrate: Si / SiO₂ + sacrificial layer



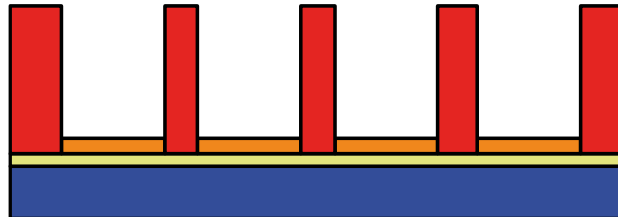
- Contact layer - B-B and A-A are connected out of the screen plane



- Thick SU-8 layer, mechanically strong and chemically resistant



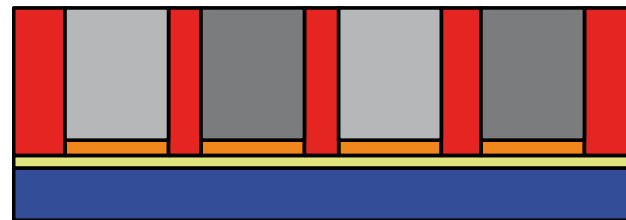
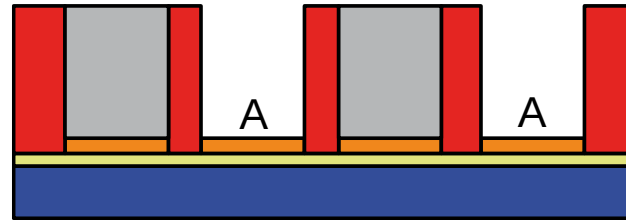
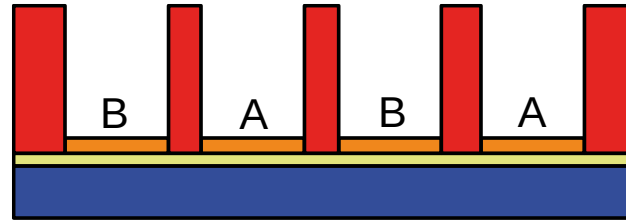
- Structuring of thermoleg template



greenTEG

Fabrication Process

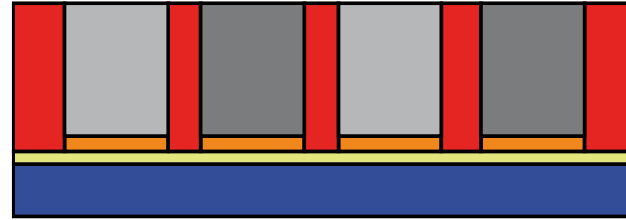
- Thermoelectric material template
- Deposition of p-type material using electrochemical deposition and contacts B
- Deposition of n-type material using electrochemical deposition and contacts A



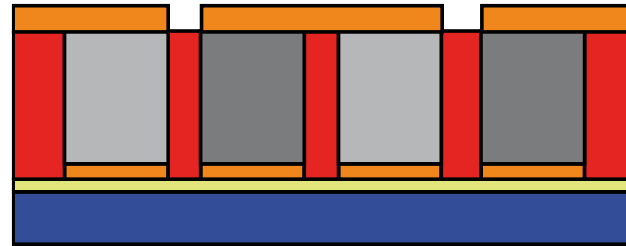
greenTEG

Fabrication Process

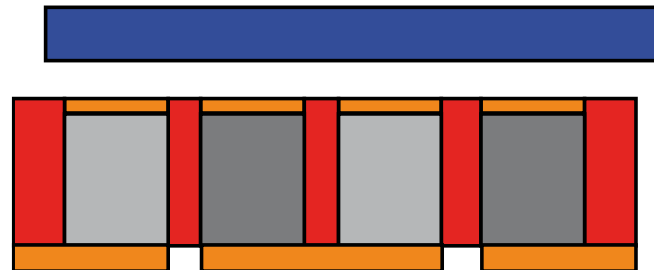
- Thermoelectric material in SU-8 template



- Deposition of top interconnects



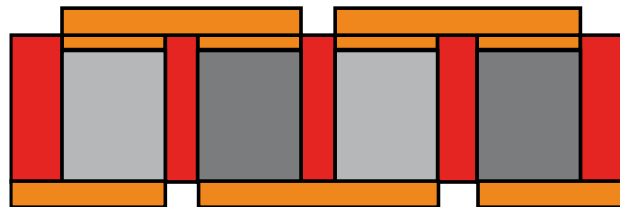
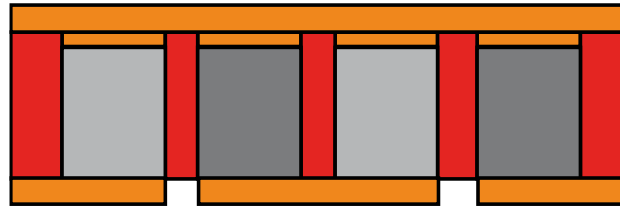
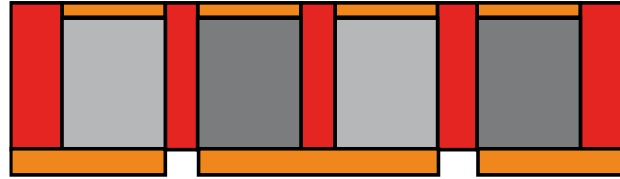
- Release of the structure by dissolution of the sacrificial layer



greenTEG

Fabrication Process

- Thermoelectric material in SU-8 template – no substrate
- Deposition of bottom (now top) interconnect layer
- Structuring of bottom contact layer and removal of connections A-A and B-B



Thermoelectric Materials

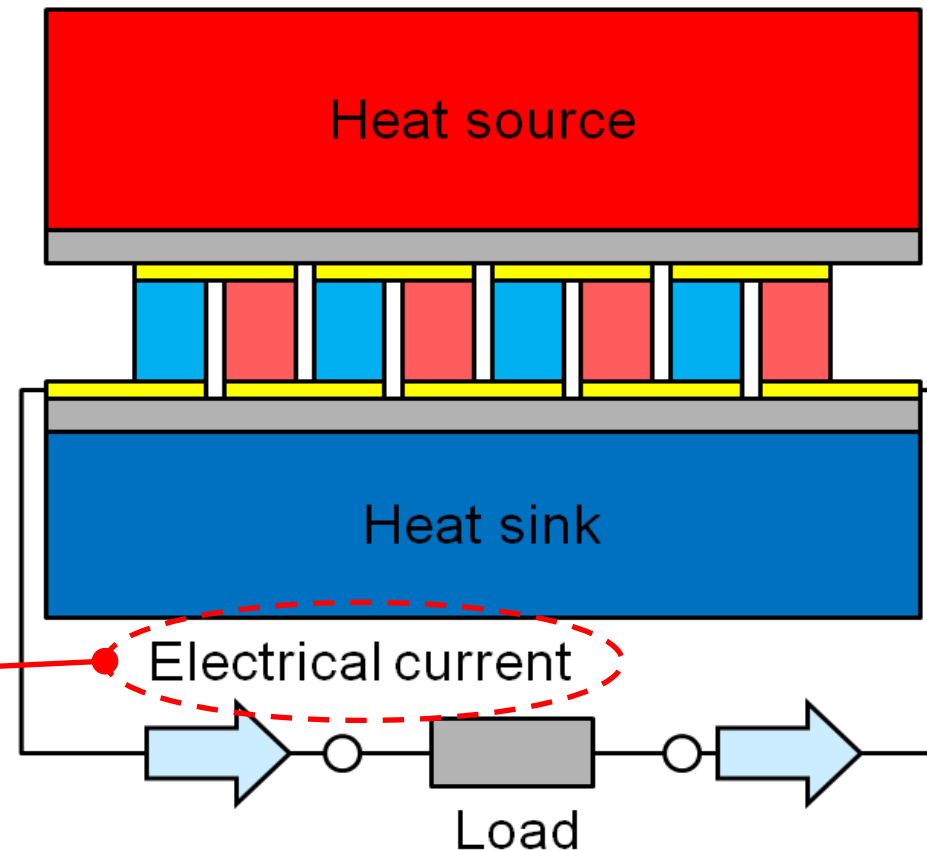
Material properties for high-performance TEGs

$$\Delta U = S \cdot \Delta T$$

- Large **Seebeck coefficient** (dependend on material choice, temperatures,...)

$$\Delta U = S \cdot \Delta T$$

- Large **thermal** resistivity
= low **thermal** conductivity
- Large **electrical** conductivity
= low **electrical** resistivity



Thermoelectric Materials

Figure of Merit

- „performance indicator“ for a thermoelectric material
- dimensionless value
- higher values of ZT indicate better material performance
- strong dependence on temperature
- only few high-ZT materials available

What does ZT mean in detail?

- high Seebeck coefficient S ➡ high output power
- high electric conductivity σ ➡ low internal resistance ➡ high output power
- low thermal conductivity κ ➡ no thermal short circuit of ΔT ➡ high output power

$$ZT = \frac{S^2 \cdot \sigma}{\kappa} \cdot T$$

S : Seebeck coefficient $\left[\frac{V}{K} \right]$

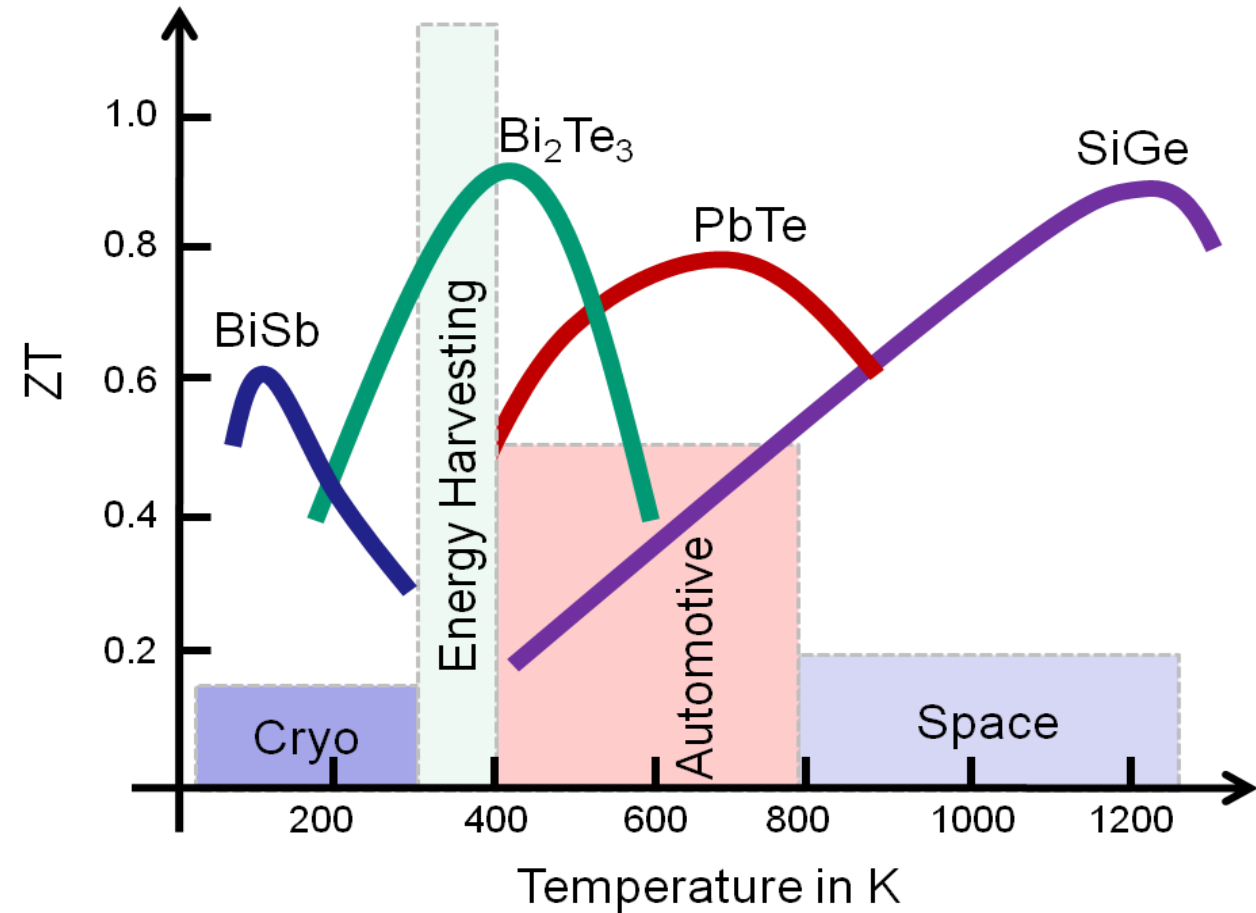
κ : thermal conductivity $\left[\frac{W}{m \cdot K} \right]$

σ : electrical conductivity $\left[\frac{1}{\Omega \cdot m} \right]$

Thermoelectric Materials

Established Materials

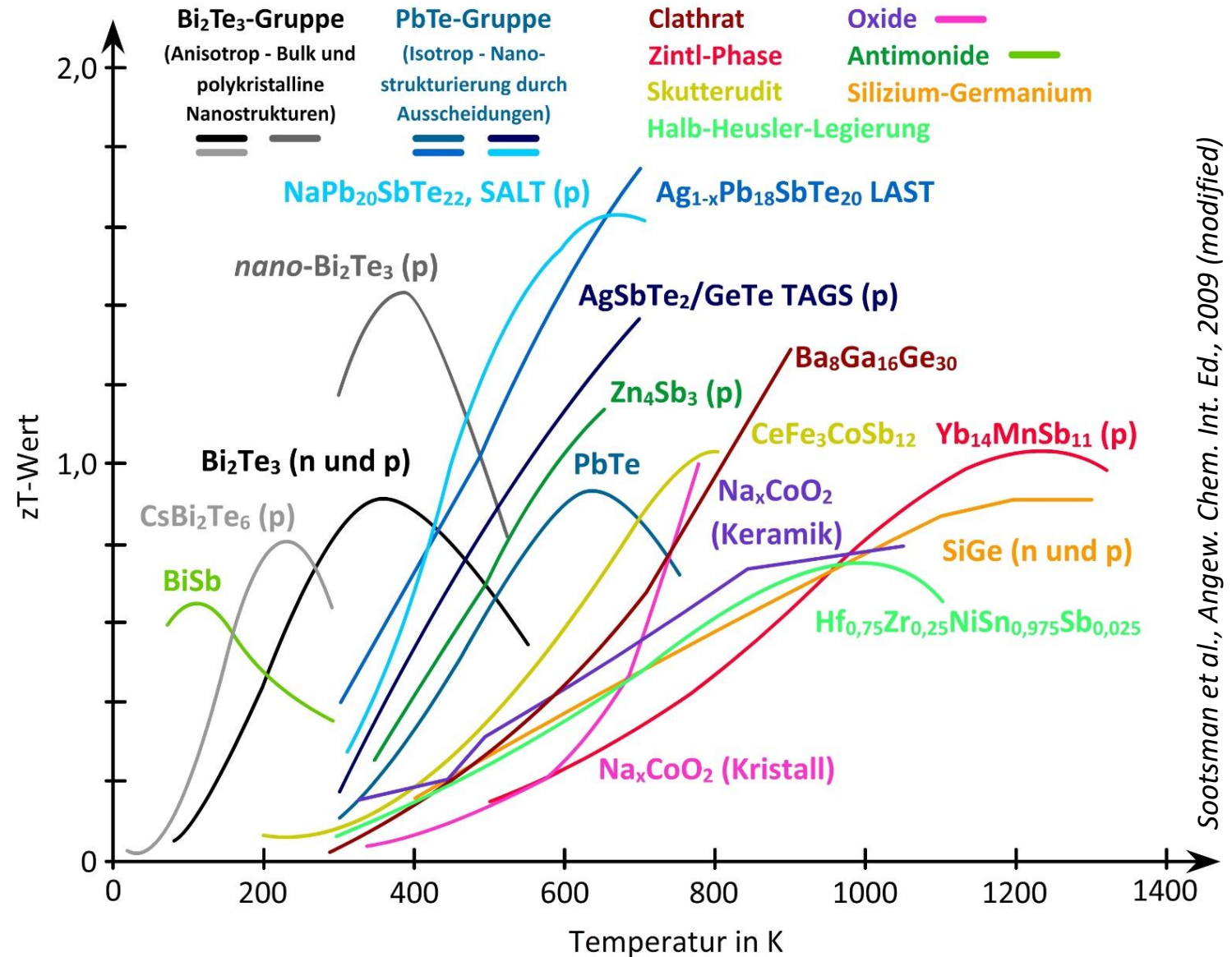
- Material choice based on temperature range and application
- ZT values of max. 1 for all materials
- Only Bi_2Te_3 commercially available
- Ongoing research for new materials:
 - Skutterudites
 - Half-Heusler
 - Clathrates



Thermoelectric Materials

Modern Materials

- Material choice based on temperature range and application
- ZT values of max. 1 for most materials
- Only Bi_2Te_3 commercially available
- Ongoing research for new materials:
 - Skutterudites
 - Half-Heusler
 - Clathrates

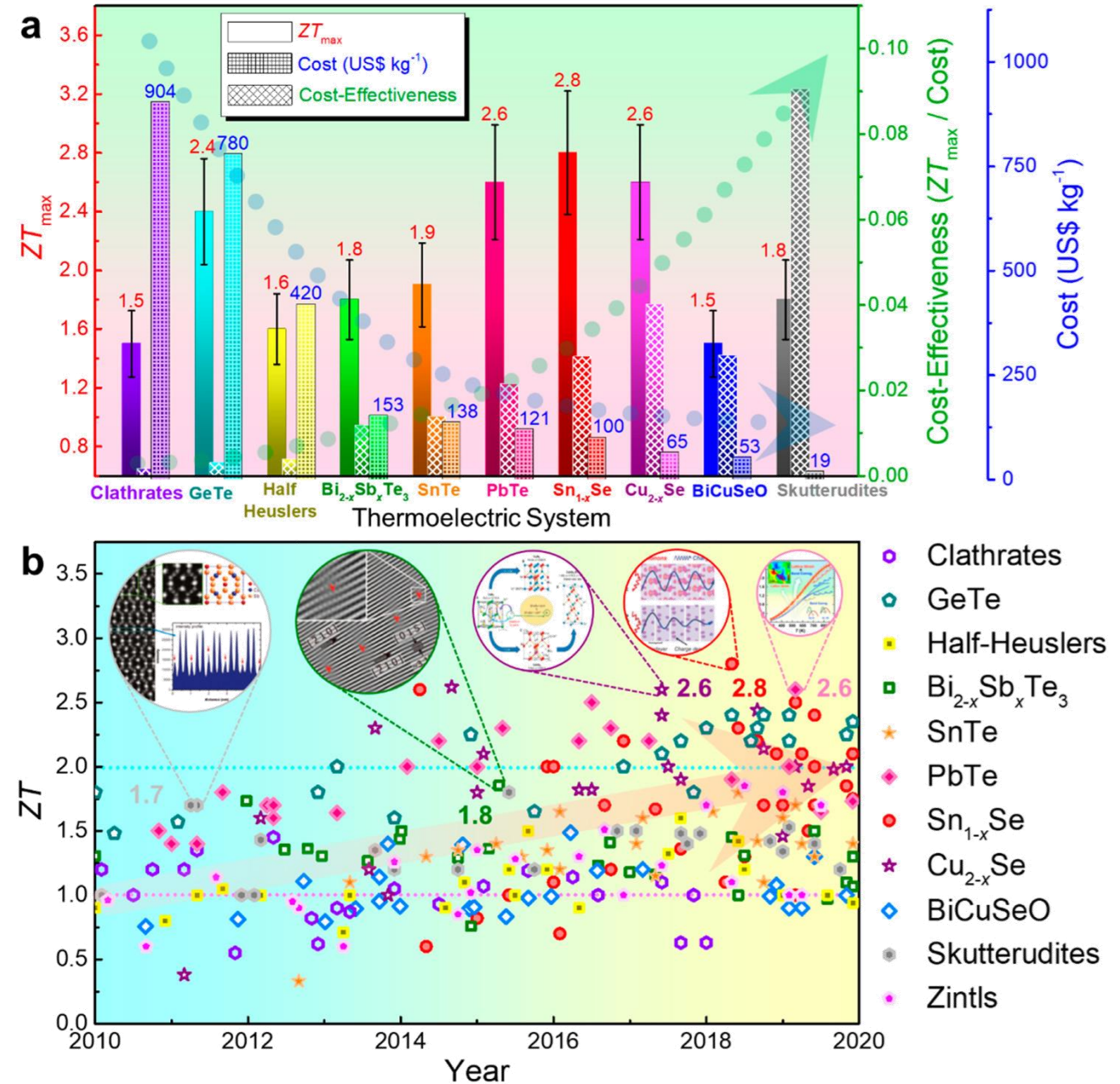


Sootsman et al., Angew. Chem. Int. Ed., 2009 (modified)

Thermoelectric Materials

Modern Materials

- Material choice based on temperature range and application
- ZT values of max. 1 for most materials
- Only Bi_2Te_3 commercially available
- Ongoing research for new materials:
 - Skutterudites
 - Half-Heusler
 - Clathrates

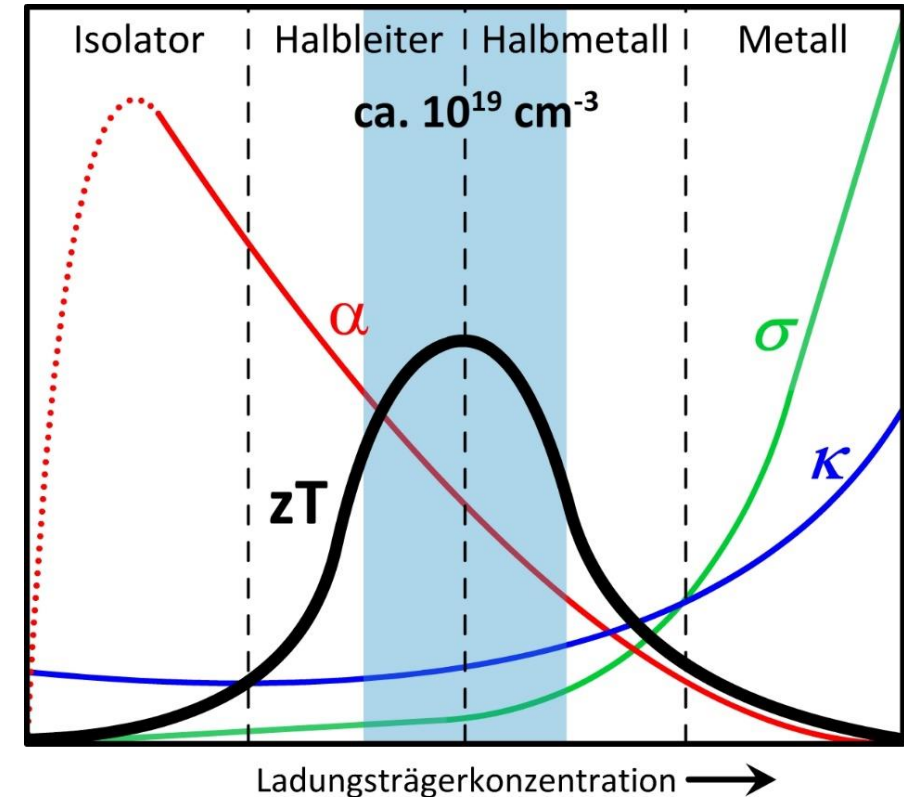


Thermoelectric Materials

ZT - Why so little Improvement?

Contradicting requirements

- S , σ and κ are not independent
 - low lattice thermal conductivity (= low κ) and high mobility (= high σ) cannot exist together
 - small bandgap degenerate p- and n-type semiconductors
 - adjustable doping levels required, not easy
 - mechanical stability required
- ➔ only few materials available



$$ZT = \frac{S^2 \cdot \sigma}{\kappa} \cdot T = \frac{PF}{\kappa} \cdot T$$

PF : "powerfactor"

$$\kappa_{ges} = \kappa_e + \kappa_L$$

Thermoelectric Efficiency

$$\eta_c = \frac{T_h - T_c}{T_h} \quad T_h: \text{hot side temperature } T_c: \text{cold side temperature}$$

Carnot limit

- Maximum efficiency of thermoelectrics is theoretically limited by Carnot
- small efficiency for small temperature differences

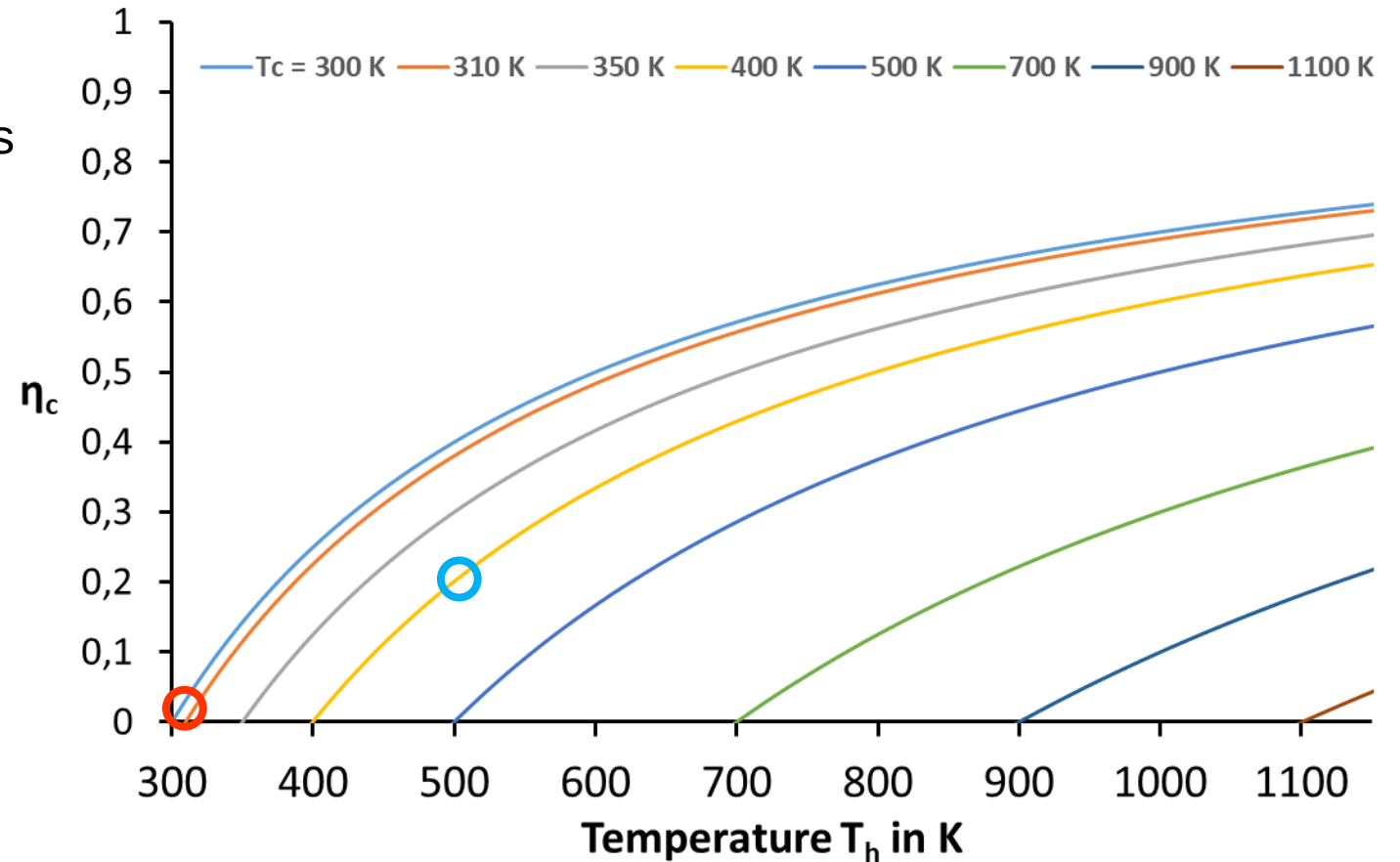
Examples:

room-temperature application

$$\left. \begin{array}{l} T_h = 310 \text{ K} \\ T_c = 300 \text{ K} \end{array} \right\} \Rightarrow \eta_c = \frac{10}{310} = 3,2\%$$

automotive application

$$\left. \begin{array}{l} T_h = 600 \text{ K} \\ T_c = 400 \text{ K} \end{array} \right\} \Rightarrow \eta_c = \frac{200}{500} = 40\%$$



Thermoelectric Efficiency

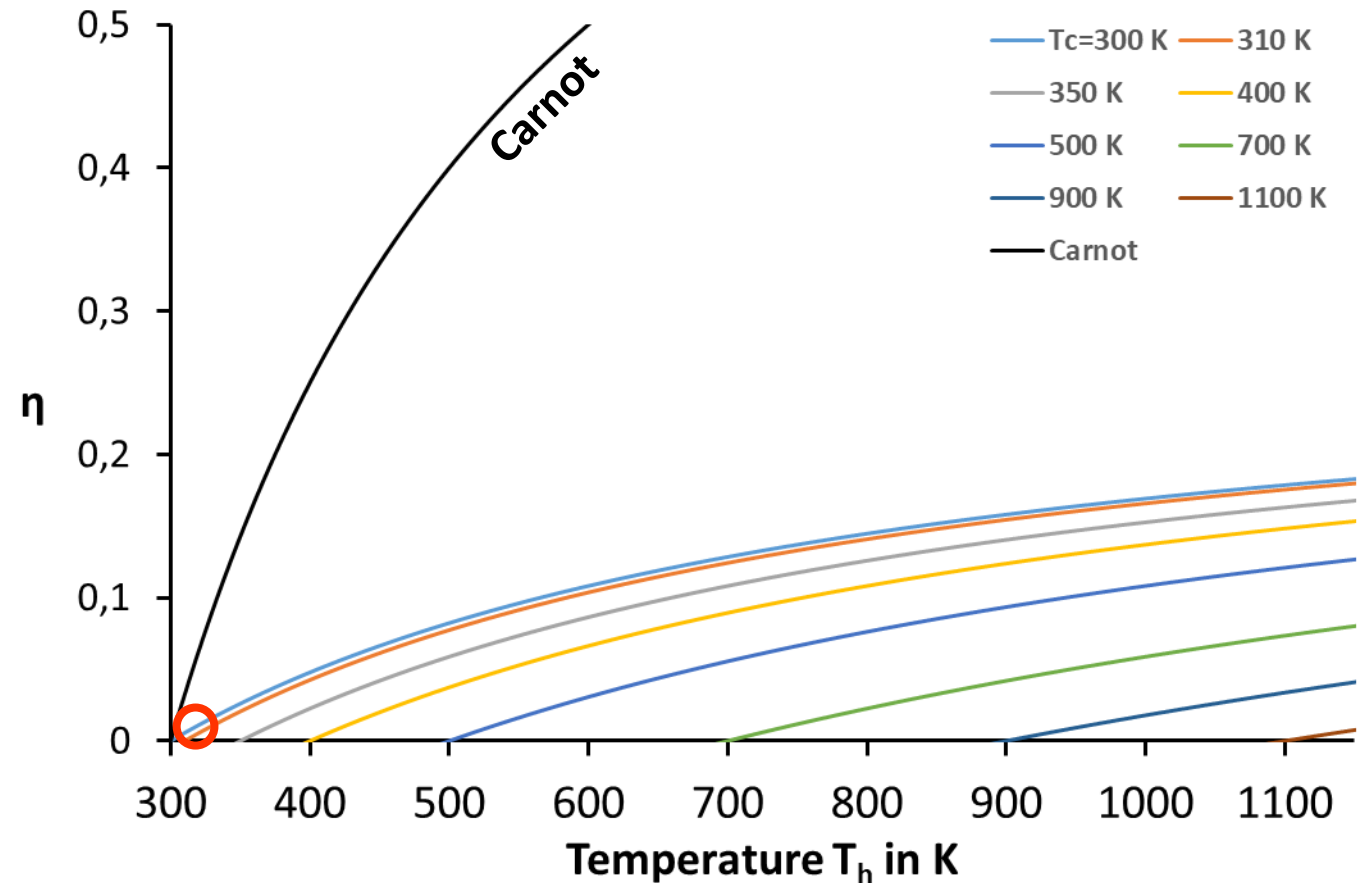
TEG Efficiency

Limits for the device efficiency

- Plot calculated assuming ZT of 1
- Best efficiency of ~ 10 % achieved in space applications with high ΔT
- The maximum efficiency of a TEG is limited by the ZT of the material

$$\eta = \frac{(T_h - T_c)}{T_h} \cdot \frac{\sqrt{ZT_m + 1} - 1}{\sqrt{ZT_m + 1} + (T_c/T_h)}$$

mit $ZT_m = \frac{\int_{T_c}^{T_h} ZT dT}{T_h - T_c}$



$$\left. \begin{array}{l} T_h = 310 \text{ K} \\ T_c = 300 \text{ K} \\ ZT = 1 \end{array} \right\} \Rightarrow \eta < \sim 0,6\%$$

Thermoelectric Efficiency

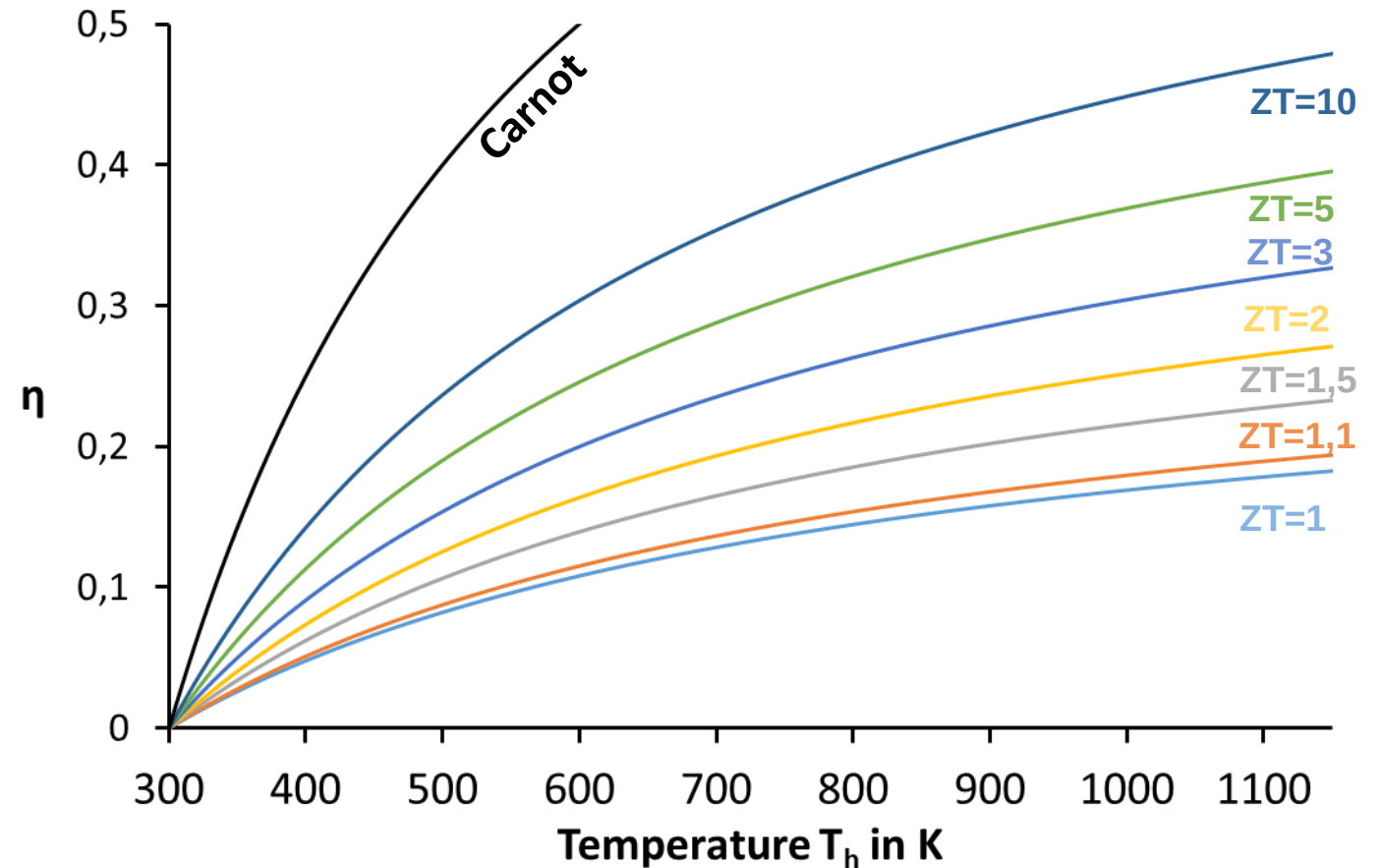
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$$\left. \begin{array}{l} T_h = 310 \text{ K} \\ T_c = 300 \text{ K} \\ ZT = 1 \end{array} \right\} \Rightarrow \eta < \sim 0,6\%$$

Power Generation

Electrical Equivalent Circuit

Voltage source with internal resistance

$$U_0 = m \cdot (S_p - S_n) \cdot (T_2 - T_1); \quad I_a = \frac{U_0}{R_{g,eff} + R_L};$$

U_0 : open circuit voltage

I_a : output current

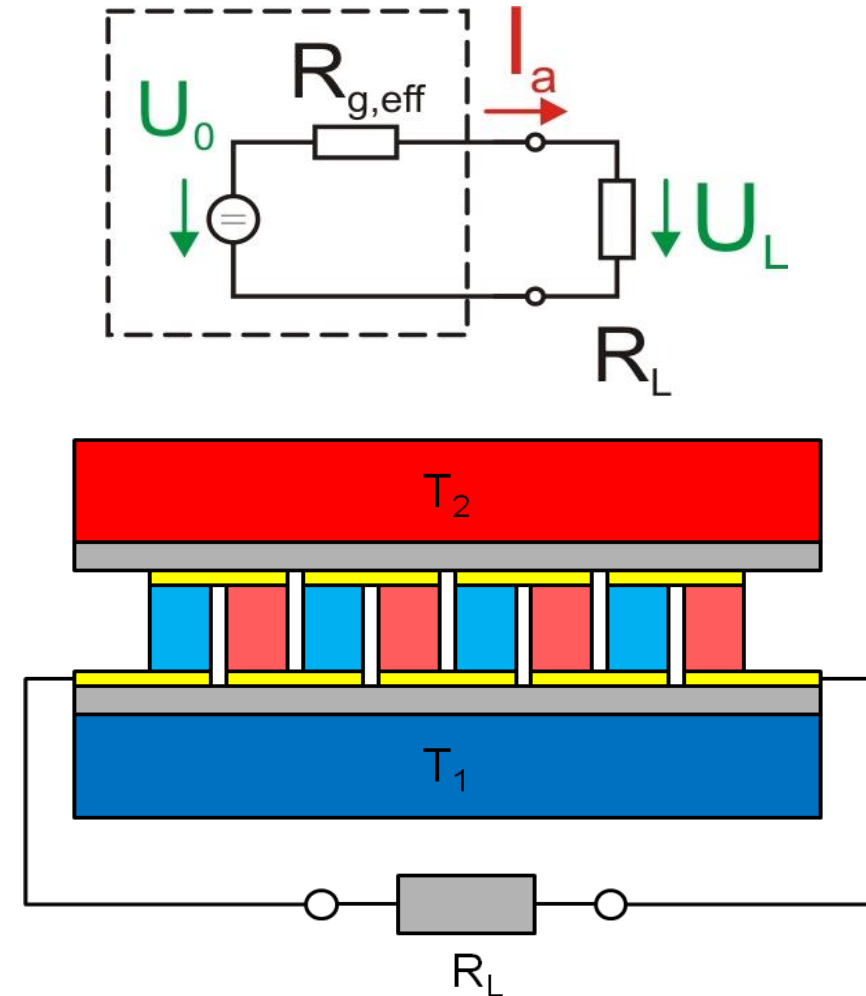
m : number of thermocouples

$T_{1,2}$: Temperatures of hot/cold side

$S_{p,n}$: Seebeck coefficient of p/n – material

$R_{g,eff}$: internal resistance of TEG

R_L : load resistance



Power Generation

Electrical Equivalent Circuit

Power across the load

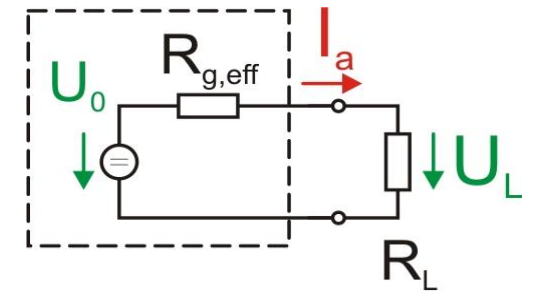
$$\left. \begin{aligned} U_O &= m \cdot (S_p - S_n) \cdot (T_2 - T_1) \\ I_a &= U_O / (R_{g,eff} + R_L) \end{aligned} \right\} \Rightarrow$$

$$P = U_L \cdot I_L = \frac{[m \cdot (S_p - S_n) \cdot (T_2 - T_1)]^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2$$

Maximum output power

$$\frac{dP(R_L)}{dR_L} \stackrel{!}{=} 0 \Rightarrow R_L = R_{g,eff} \Rightarrow$$

$$P_{max} = \frac{U_0^2}{4 \cdot R_{g,eff}}; \text{ for } R_L = R_{g,eff}$$



Power Generation Evaluation

Influence factors on the electric output power

- higher Seebeck coefficients ➡ higher output power ++
- higher temperature difference $T_2 - T_1$ ➡ higher output power ++
- higher number of thermocouples m ➡ higher output voltage ++
➡ higher internal resistance -- 

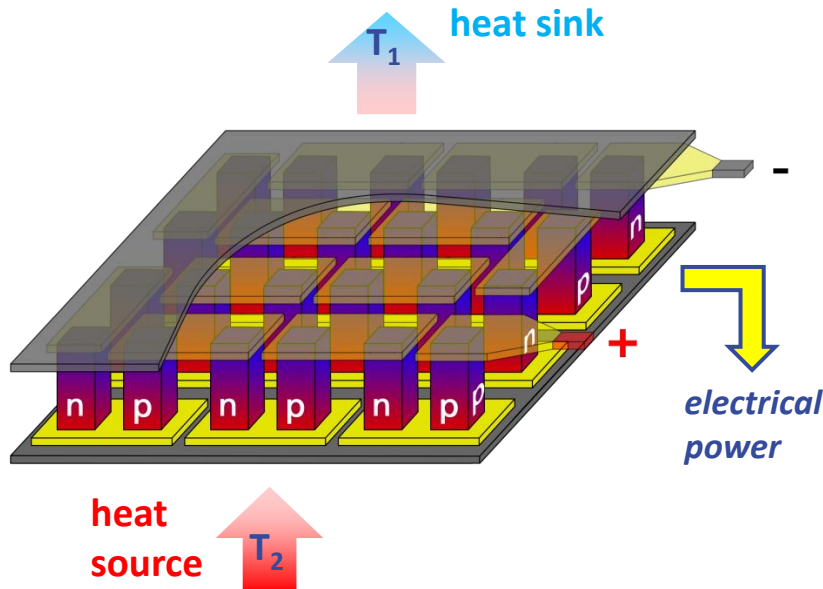
$$P = U_L \cdot I_L = \frac{[m \cdot (S_p - S_n) \cdot (T_2 - T_1)]^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2$$

Result

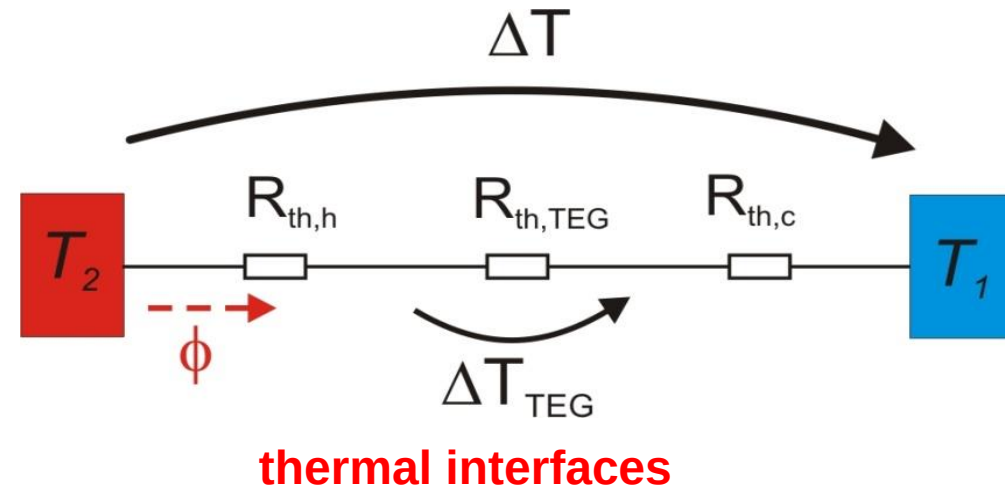
- Higher Seebeck coefficients S_p and S_n (i.e. better materials) are advantageous.
- Higher temperature differences ΔT will increase the output power: $P \sim (\Delta T)^2$
- Increasing of the number of thermocouples m generates a **design conflict**. 

TEG Modeling

Thermal Interfaces



Result

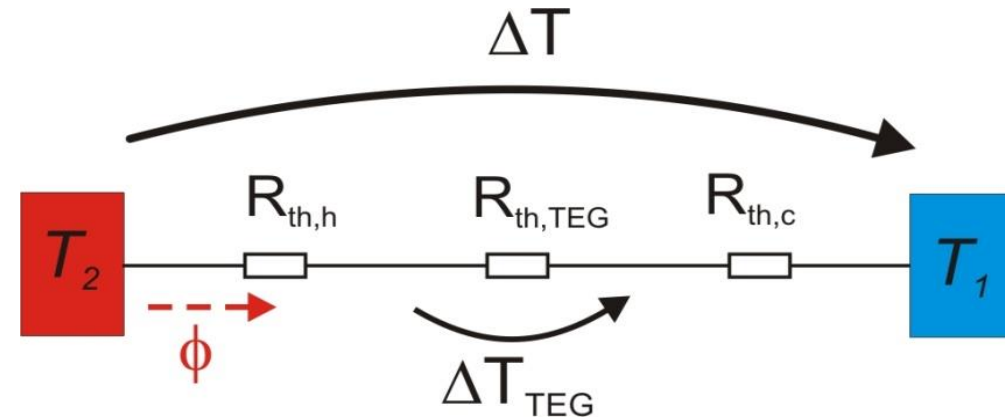
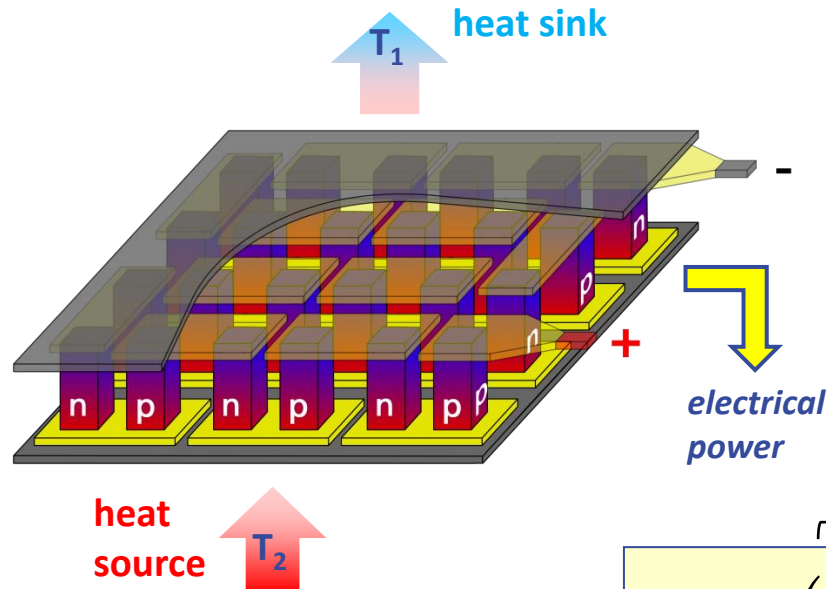


$$\Delta T_{TEG} = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,h} + R_{th,c}} \right) \cdot (T_2 - T_1)$$

- The thermal resistances between the TEG and the hot/cold side will reduce the effective temperature ΔT_{TEG} , i.e. the thermal input power
 - Further problems from thermal expansion of different materials
- ➡ A stable thermal coupling with low thermal resistances is essential!

TEG Modeling

Thermal and Electrical Interfaces



thermal interfaces

$$P_{el} = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,h} + R_{th,c}} \right)^2 \cdot \frac{(m \cdot S \cdot \Delta T)^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2$$

TEG

electrical interface

Result

- Thermal interfaces reduce the thermal input power
- Adaptation of the load resistance is mandatory

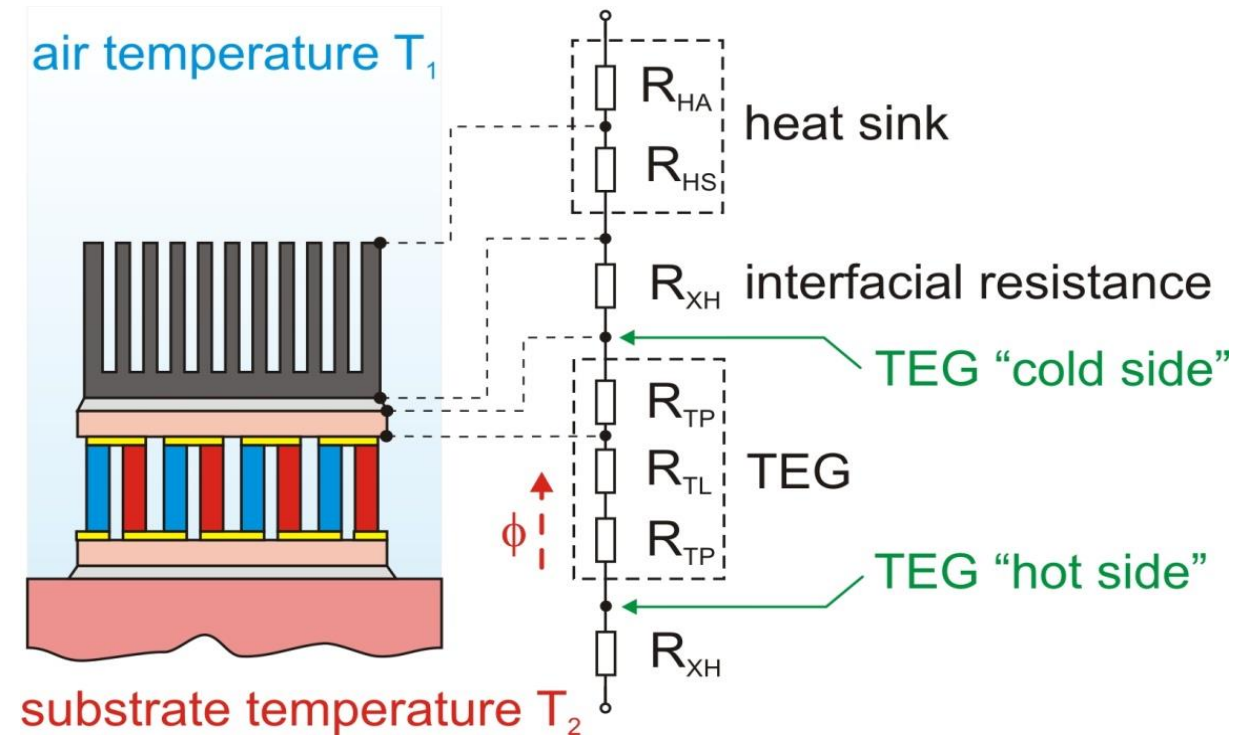
TEG Modeling

Thermal Interfaces

Example

A TEG is coupled to the ambient air via a heat sink. We consider the heat flux to the „cold side“.

Type	Location	Name
internal heat resistance	TEG, heat sink	R_{TP} , R_{TL} , R_{HS}
contact resistance or interfacial heat resistance between solid materials	TEG - heat sink	R_{XH}
heat resistance to air or liquid	heat sink - air	R_{HA}

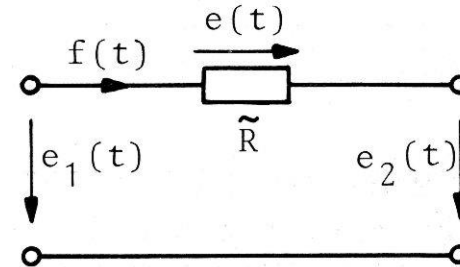
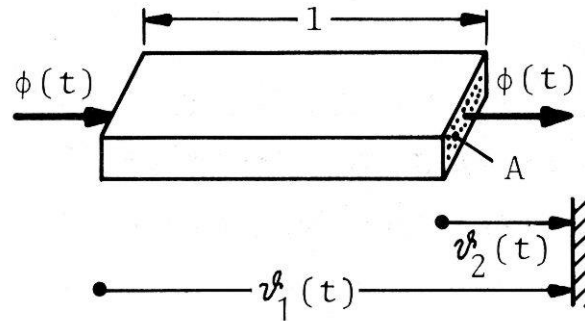


Thermal Interfaces

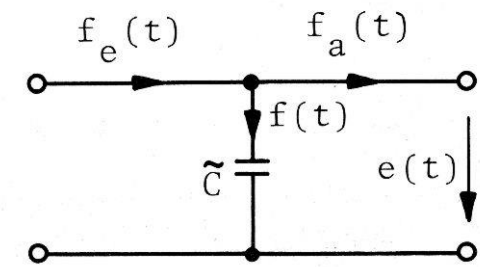
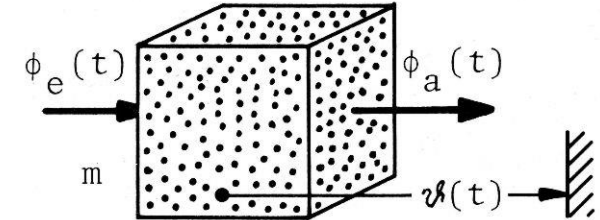
Thermal Elements and their Electrical Analog

1. Thermal potential and flow

- Potential $e(t) \rightarrow$ temperature $T(t)$
- Current / flow $f(t) \rightarrow$ heat flow $\phi(t)$



2. Relevant thermal elements



$$\text{heat resistance: } R_{th} = \frac{l}{\lambda \cdot A} \left[\frac{K}{W} \right]$$

$$\lambda: \text{ thermal conductivity } \left[\frac{W}{m \cdot K} \right]$$

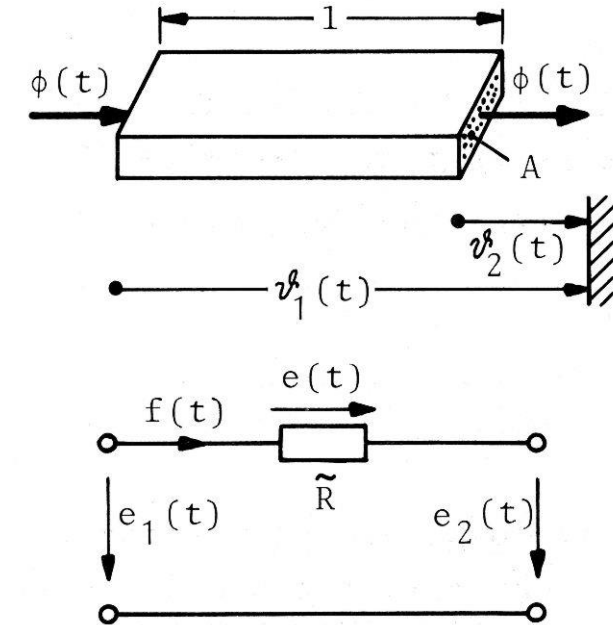
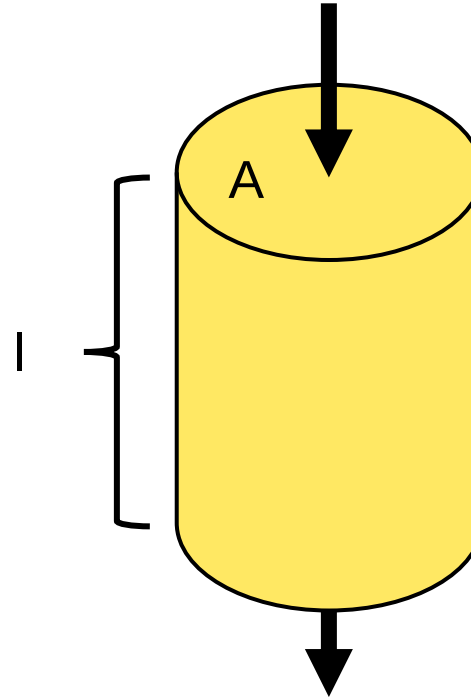
$$\text{thermal capacitance: } C = m \cdot c \left[\frac{J}{K} \right]$$

$$c: \text{ specific heat capacitance } \left[\frac{J}{kg \cdot K} \right]$$

Thermal Resistances

Volumetric Definition

- best for 3D bodies
- realistic assumption of a one-dimensional heat flow
- area-specific thermal conductivity λ as material parameter



Limitations

- no representation of anisotropic heat conductivity
- no representation of 2D or 3D heat flow

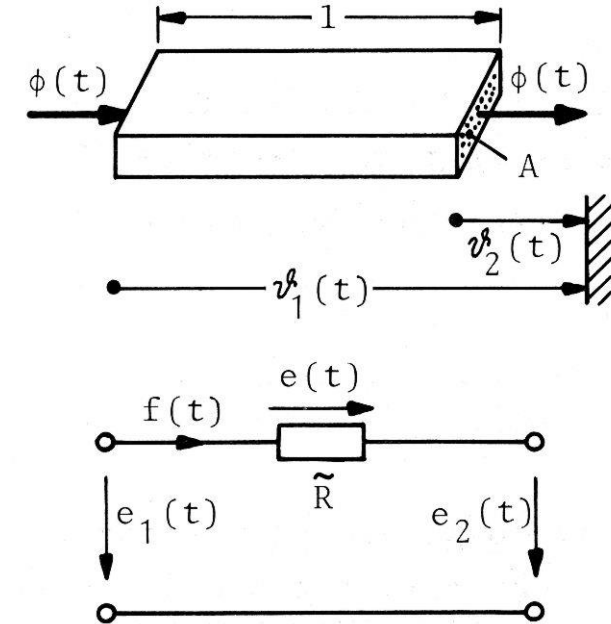
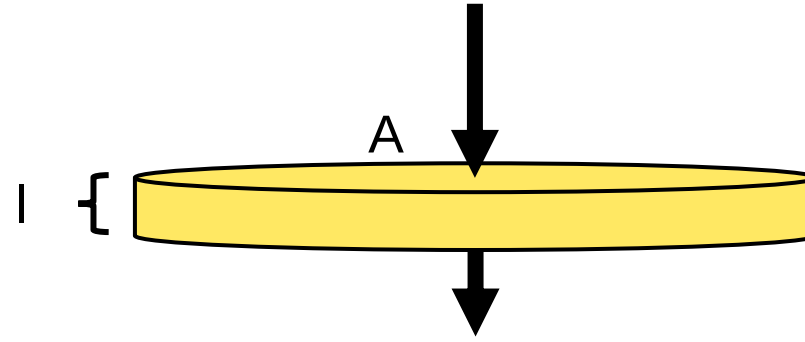
$$\text{heat resistance: } R_{th} = \frac{l}{\lambda \cdot A} \left[\frac{K}{W} \right]$$

$$\lambda: \text{ thermal conductivity } \left[\frac{W}{m \cdot K} \right]$$

Thermal Resistances

Area-specific Definition

- best for flat (2D-like) bodies and contact areas ($l \ll \sqrt{A}$)
- realistic assumption of a one-dimensional heat flow
- area-specific heat resistance ρ_{th} or contact resistance γ as material parameters



Limitations

- anisotropic heat conductivity can be included in ρ_{th}
- layer thickness l has to be known
- no representation of 2D or 3D heat flow (usually not required)

$$R_{XH} = \frac{\gamma}{A}; \quad R_{XY} = \frac{\rho_{th}}{A};$$

γ : specific contact resistance
 ρ_{th} : specific heat resistance
 A : effective area

Thermal Resistances

Area-specific Definition

Some values for contact and thin-layer heat resistances

Specific heat resistance

Material	Thickness	ρ_{th} in K·cm ² /W
Glimmer	50 µm	1.40
Silicone	300 µm	2.46
Al ₂ O ₃	300 µm	0.12
Polyimide	77 µm	1.71
Silicone-TCP	100 µm	1.64
WL-Kleber (Epoxy)	100 µm	1.22

Source: Fischer elektronik

$$R_{XH} = \frac{\gamma}{A}; \quad R_{XY} = \frac{\rho_{th}}{A};$$

γ : specific contact resistance
 ρ_{th} : specific heat resistance
 A : effective area

Specific contact resistance

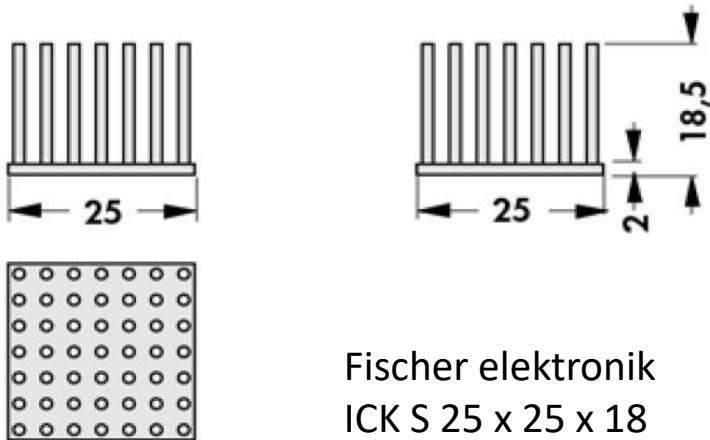
γ in cm ² ·K/W	with thermally conductive paste (TCP)	without
metal on metal	0.5	1
metal on eloxal	1.4	2

Source: Nührmann, Werkbuch Elektronik, Franzis, 1980

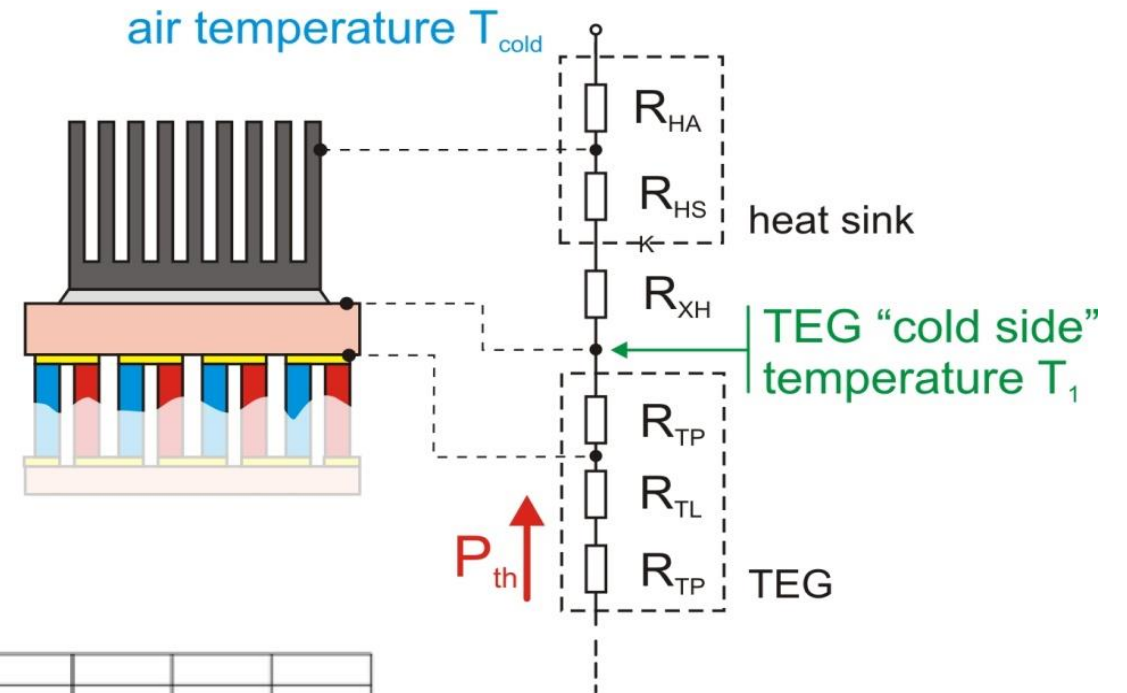
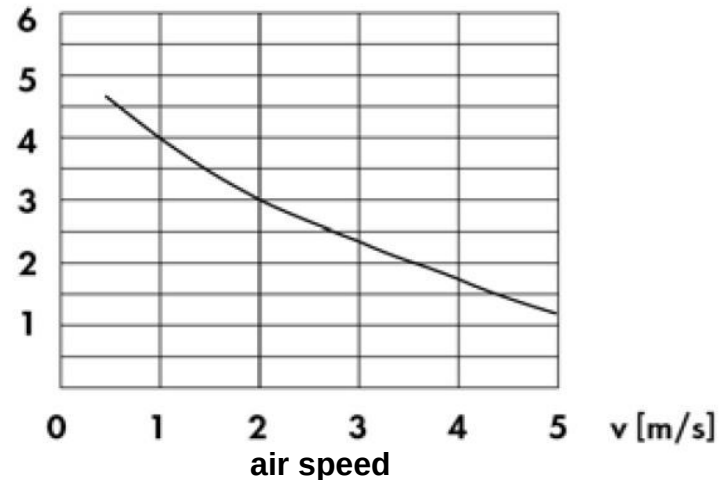
Thermal Interfaces

Influence of Air Convection

- variable heat resistance R_{HA} at the heat sink - air interface
- significant increase of R_{HA} without enforced convection (air speed $v \rightarrow 0$)



R_{th}
[K/W]

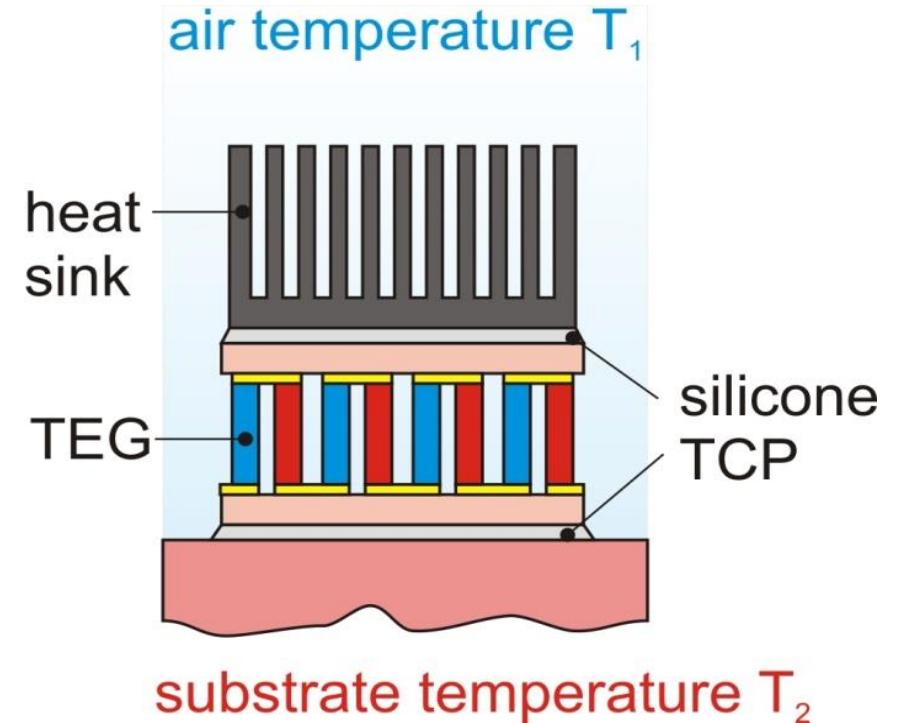


TEG Power Output Calculation

Parameters

Example calculation for an air-cooled TEG

- **TEG: thermalforce TEG 049-150-30**
area: 25 x 25 mm²
heat resistance: 5.263 K/W
- **Heat sink: Fischer elektronik ICK S 25 x 25 x 18**
area: 25 x 25 mm²
heat resistance: 4.5 K/W (low air convection)
1.5 K/W (forced air convection)
- **Interface: thermally conductive paste (TCP, Wacker P12)**
layer thickness: 50 µm
thermal conductivity: $\lambda = 0.81 \text{ W/(mK)}$
- **hot/cold side:** $T_2 = 283 \text{ K}$; $T_1 = 273 \text{ K}$; $\Delta T = T_2 - T_1 = 10 \text{ K}$



TEG Power Output Calculation

Thermal Interfaces only

$$\Delta T_{TEG} = \underbrace{\left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,h} + R_{th,c}} \right)}_{\text{thermal feed factor } d} \cdot (T_2 - T_1)$$

- Case 1: low air convection

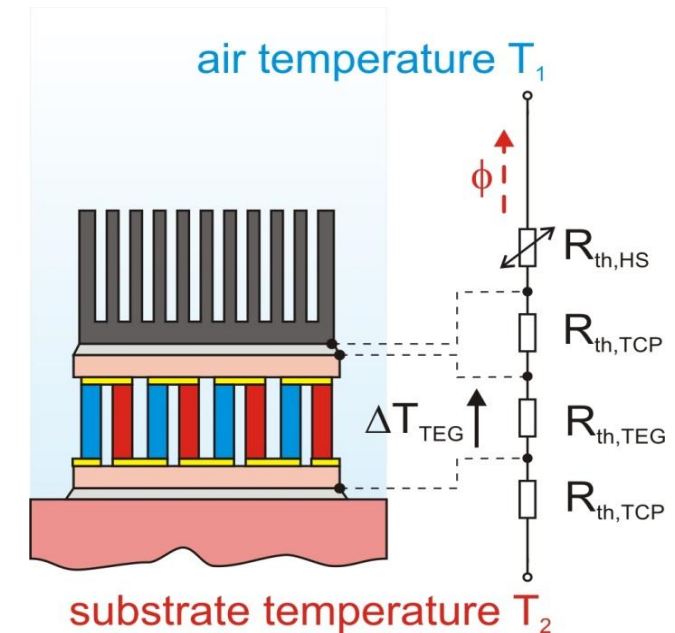
$$\left. \begin{aligned} R_{th,TEG} &= 5.263 \text{ K/W} \\ R_{th,HS} &= 4.5 \text{ K/W} \\ R_{th,TCP} &= 0.1 \text{ K/W} \end{aligned} \right\} \Rightarrow$$

$$\begin{aligned} \Delta T_{TEG} &= \underbrace{0.528}_d \cdot (T_2 - T_1) \\ P_{thermal} &= \underbrace{0.278}_{d^2} \cdot P_{max} \end{aligned}$$

- Case 2: forced air convection

$$\left. \begin{aligned} R_{th,TEG} &= 5.263 \text{ K/W} \\ R_{th,HS} &= 1.5 \text{ K/W} \\ R_{th,TCP} &= 0.1 \text{ K/W} \end{aligned} \right\} \Rightarrow$$

$$\begin{aligned} \Delta T_{TEG} &= \underbrace{0.759}_d \cdot (T_2 - T_1) \\ P_{thermal} &= \underbrace{0.576}_{d^2} \cdot P_{max} \end{aligned}$$



Result

- small **thermal feed factors** d
- power reduction proportional to d^2

TEG Power Output Calculation

Thermal and Electrical Interfaces

thermalforce TEG 049-150-30

- internal resistance $R_{g,eff}$: 0.57 Ohm
- Seebeck coefficient $m \cdot S$: 0.016 V/K
- Case 1: low air convection

$$\left. \begin{array}{l} d = 0.528 \\ \Delta T = 10 \text{ K} \\ R_L = R_{g,eff} \end{array} \right\} \Rightarrow \begin{array}{l} U_{0,real} = 84.4 \text{ mV} \\ P_{real} = 3.1 \text{ mW} \end{array}$$

- Case 2: forced air convection

$$\left. \begin{array}{l} d = 0.759 \\ \Delta T = 10 \text{ K} \\ R_L = R_{g,eff} \end{array} \right\} \Rightarrow \begin{array}{l} U_{0,real} = 121.4 \text{ mV} \\ P_{real} = 6.4 \text{ mW} \end{array}$$

$$P_{el} = d^2 \cdot \frac{(m \cdot S \cdot \Delta T)^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2$$

$$\begin{array}{l} U_{0,max} = m \cdot S \cdot \Delta T; \quad U_{0,real} = d \cdot m \cdot S \cdot \Delta T \\ P_{max} = \frac{1}{4} \cdot \frac{U_{0,max}^2}{R_L}; \quad P_{real} = d^2 \cdot P_{max} \end{array}$$

➔ @ 10 K ΔT : $U_{0,max} = 160 \text{ mV}$, $P_{max} = 11.2 \text{ mW}$

Result

- output voltage: $U_{0,real} \sim d$
- output power: $P_{real} \sim d^2$
- significant influence of thermal interfaces!

TEG Power Output Calculation

Power Losses

$$P_{el} = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,h} + R_{th,c}} \right)^2 \cdot \frac{(m \cdot S \cdot \Delta T)^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2 \cdot \underbrace{\eta_{DC-DC}}_{\text{efficiency of an electronic step-up converter}}$$

Low output voltages (+100 mV) are not sufficient to power up an electronic system. A low-voltage step-up converter is required to boost the TEG output voltage up to an acceptable level (e.g. 2 V)

**efficiency of
an electronic
step-up converter**

Power losses in the step-up converter

- low efficiency at low input voltages (typically 25 % or less)
- minimum start-up voltage required (today mostly ~ 20 mV, some ~ 10 mV)
 - ➡ consequence: No harvesting at „really low temperature gradients“

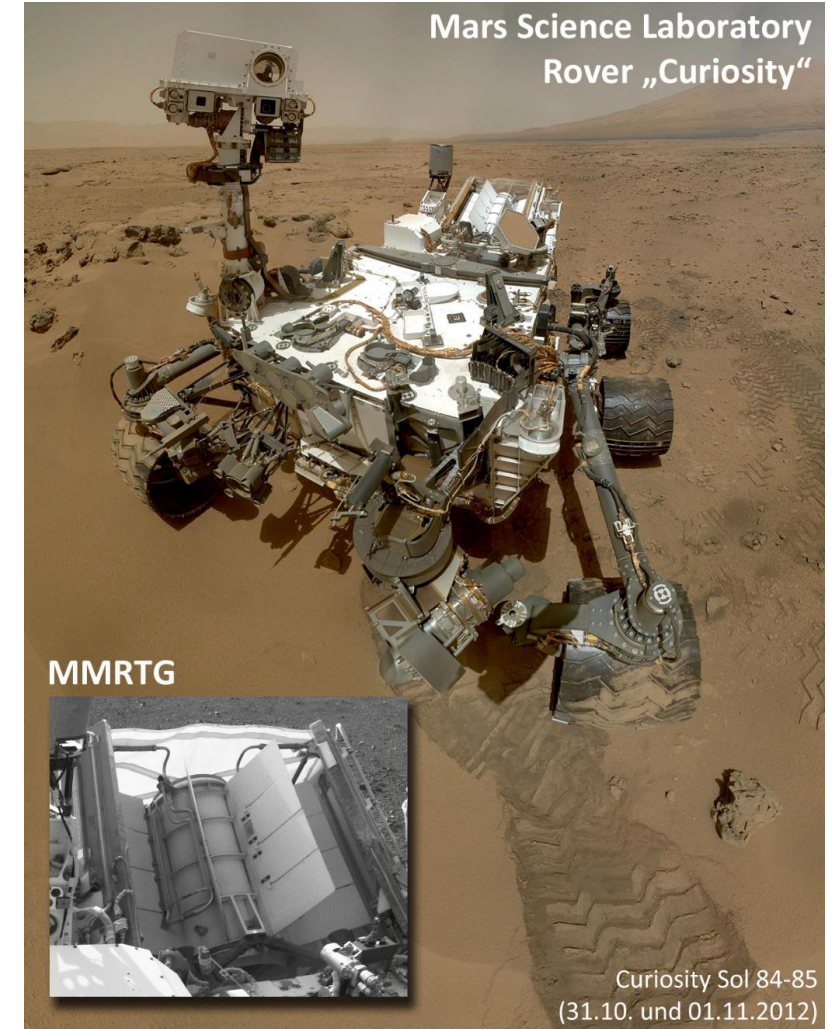
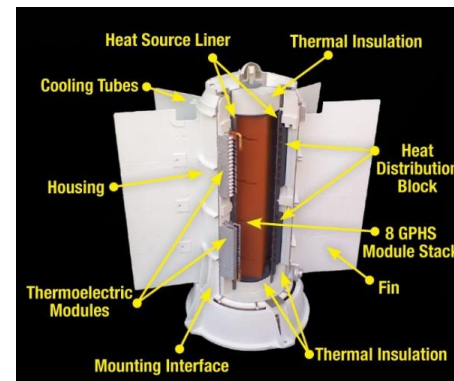
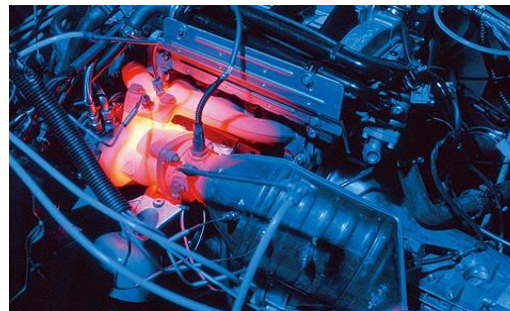
Thermoelectric Energy Harvesting Applications

- Space
 - RTGs as power supply for deep space missions with no solar radiation
 - Radioactive decay as heat source
 - High reliability (40 years +)
- Automotive
 - Power generation from exhaust heat
 - Possible alternator replacement?



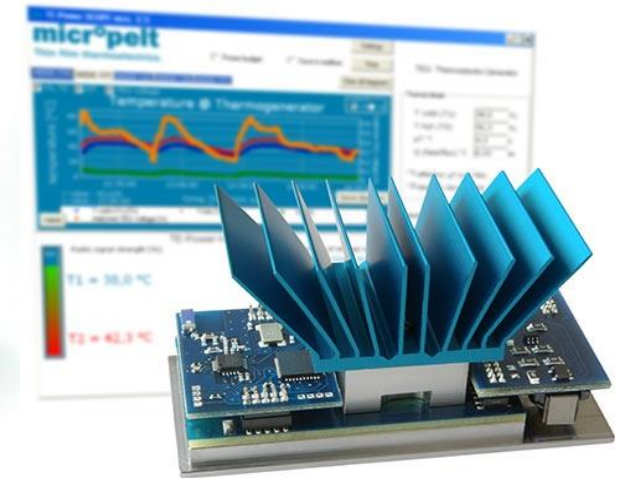
Images by NASA/JPL

Image by BMW

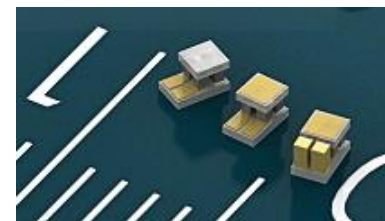
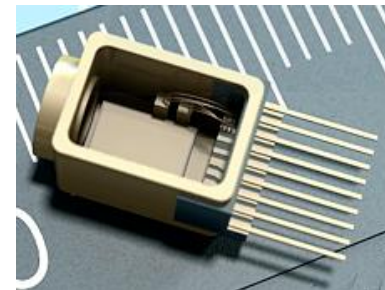


Thermoelectric Energy Harvesting & Cooling Applications

- Sensors and small scale integration
 - Wireless sensor nodes and networks
 - Independent of power supply / battery
 - Small devices possible through MEMS technology
 - Rugged environments
- Micro-scale Cooling
 - Laser wavelength stabilization in optical fiber communication systems
 - Medical and aerospace applications



Images by micropelt



Images by RMT Ltd.

Thermoelectric Cooling Applications

- Cooling boxes
 - Most common use of Peltier elements
 - Low efficiency, but easy to design
 - Space efficient and “cheap”
- Laboratory Equipment e.g. PCR Thermal Cycler
 - Precise temperature control ± 0.25 K accuracy
 - TEG robust and long-lasting



Images by Ezetil



Images by ThermoFischer

Conclusion

- **Thermoelectric energy harvesting** utilizes the Seebeck-Effect to convert a thermal flow into a DC voltage
- **Fabrication technologies** range from soldered μ TEGs to sputter-deposition and galvanic deposition to form thin layers of thermoelectric materials
- **Thermoelectric efficiency** depends on the materials Seebeck coefficient, as well as electrical and thermal conductivities and is always a compromise due to contradictory requirements
- **TEG power output** is maximized when internal and external loads are matched and increases with temperature difference $P \sim (\Delta T)^2$
- **ALL thermal interfaces** between heat source and heat sink need to be taken into account for calculations on TE-harvesting systems
- **Removing the heat from the heat sink** to sustain the temperature gradient provides the toughest challenge in designing TE-harvesting systems

$$ZT = \frac{S^2 \cdot \sigma}{\kappa} \cdot T$$